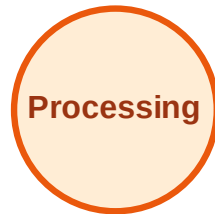


BFS Traversal

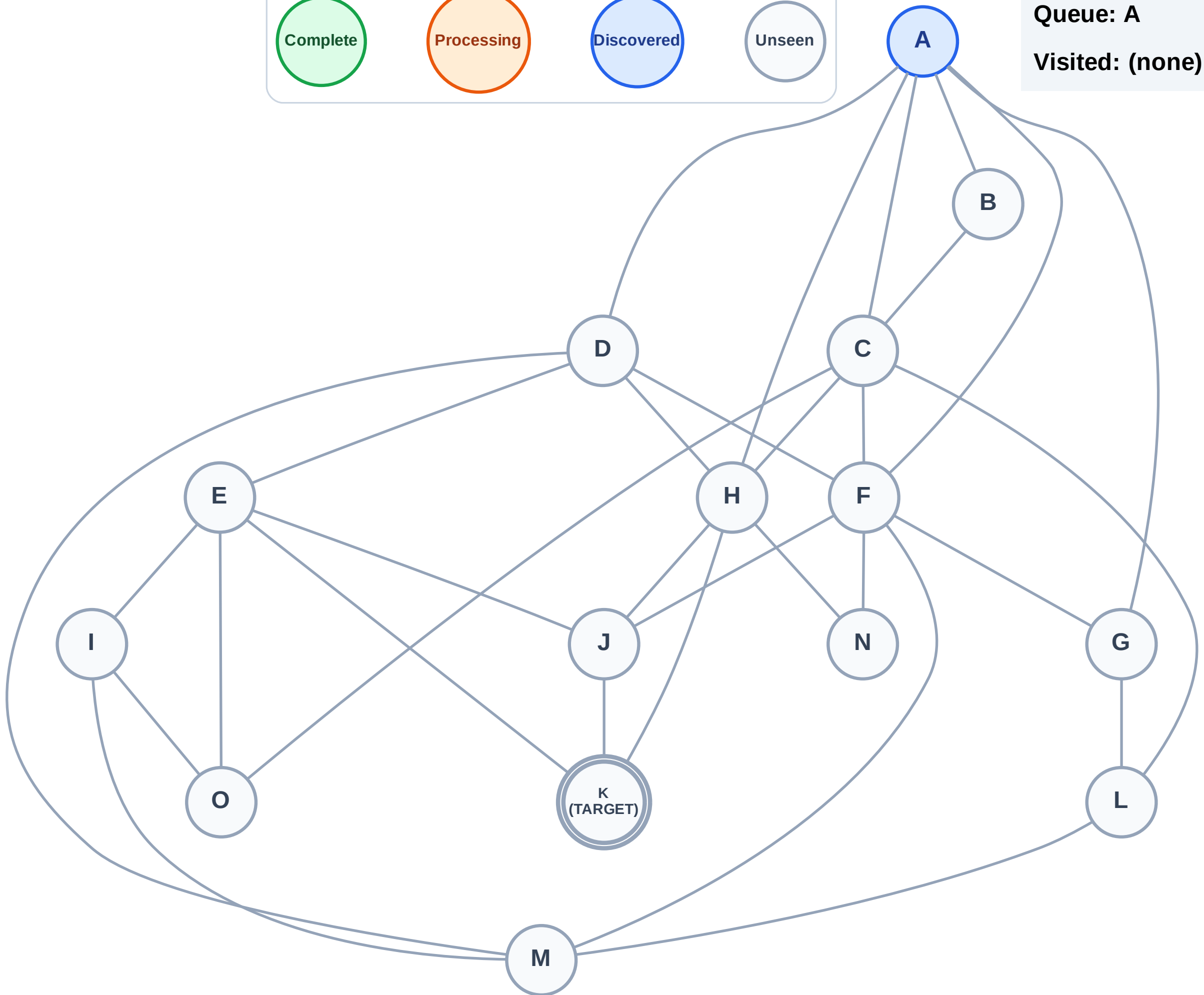
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

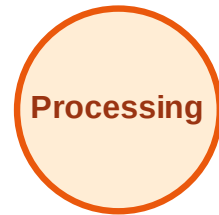
Visited: (none)



BFS Traversal

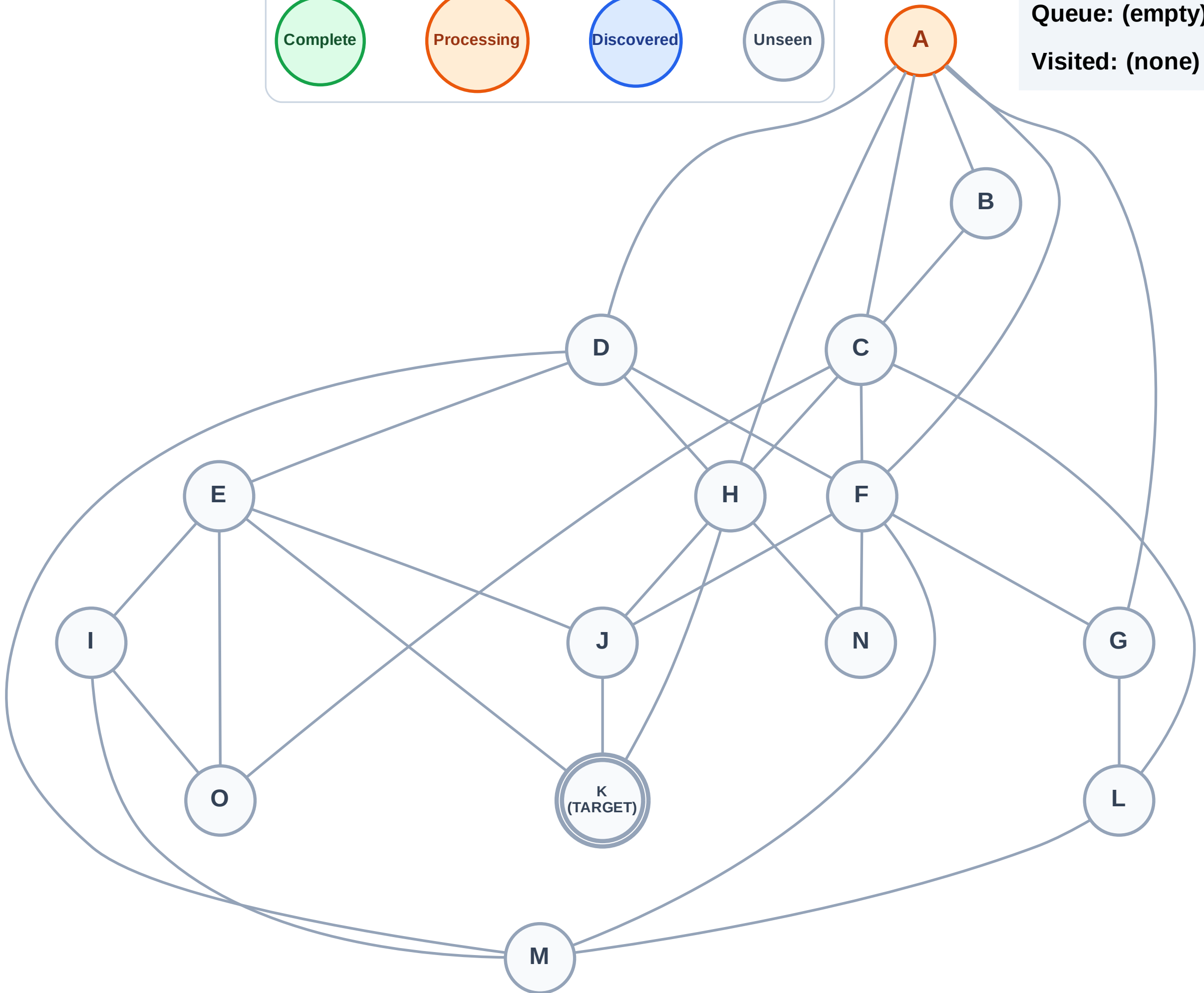
Step 2: Process node A

Legend



Queue: (empty)

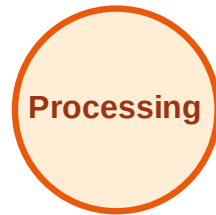
Visited: (none)



BFS Traversal

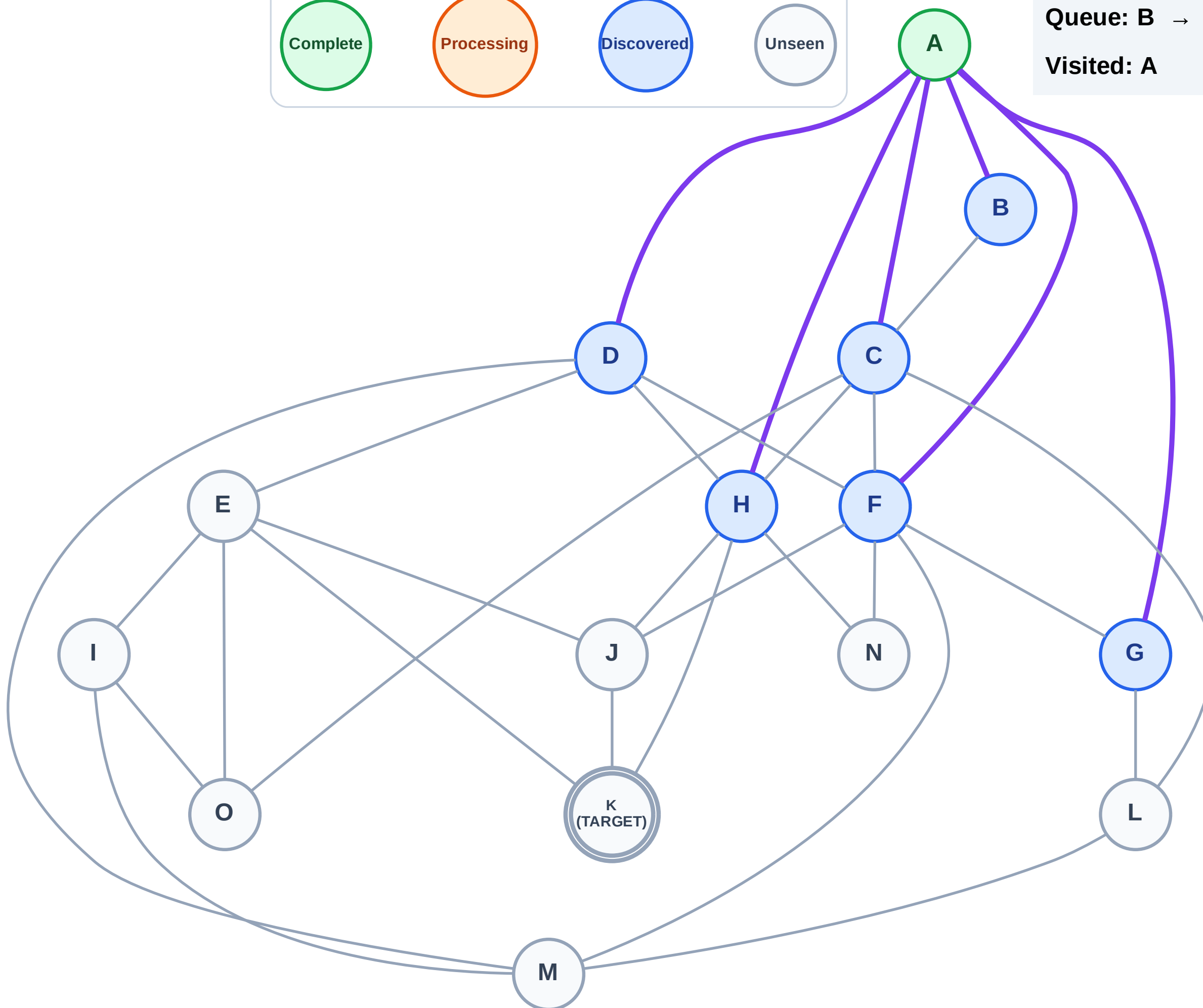
Step 3: Discover B, C, D, F, G, H from A

Legend



Queue: B → C → D → F → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

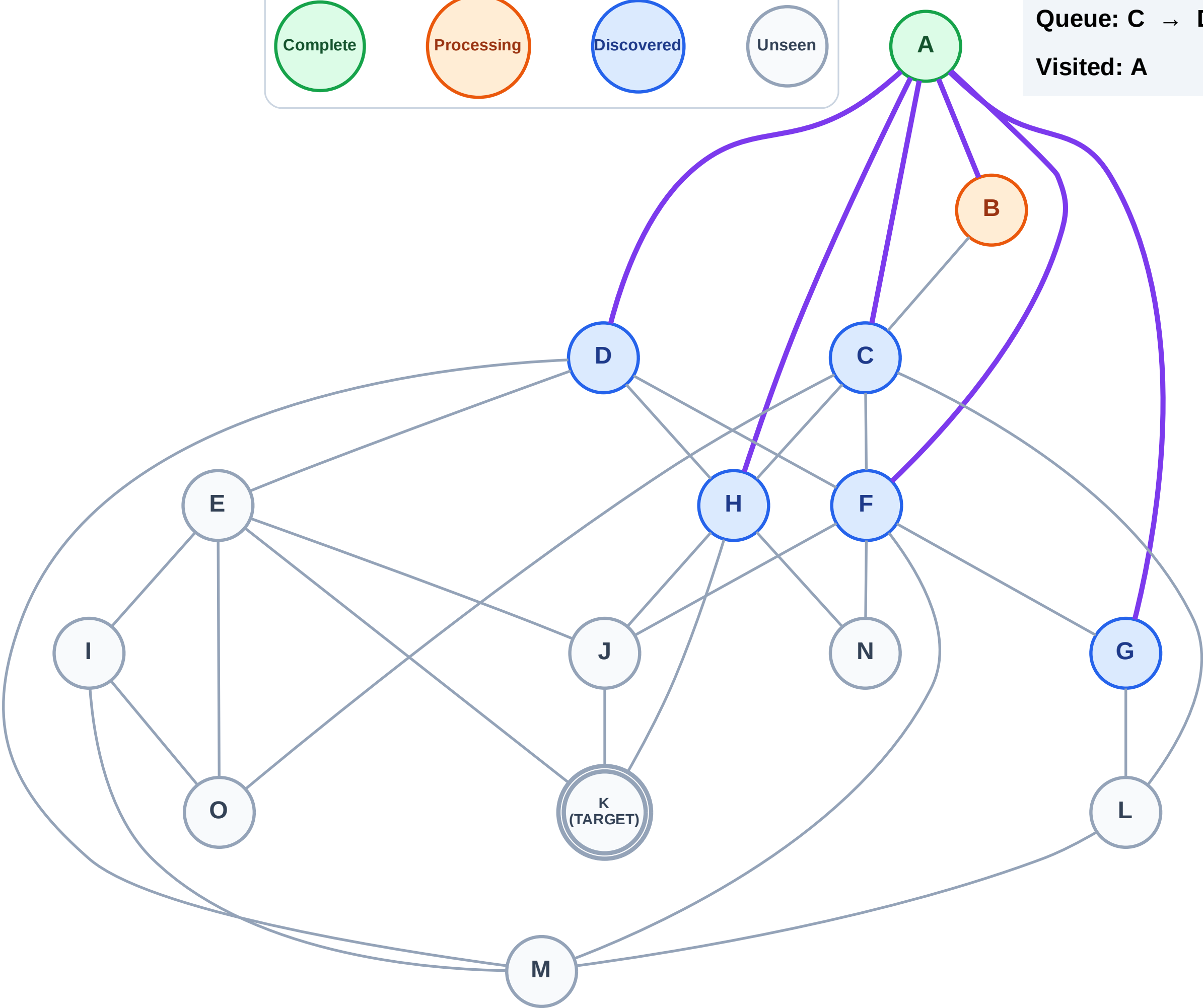
Processing

Discovered

Unseen

Queue: C → D → F → G → H

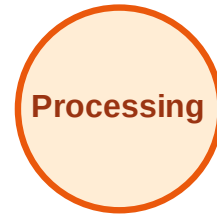
Visited: A



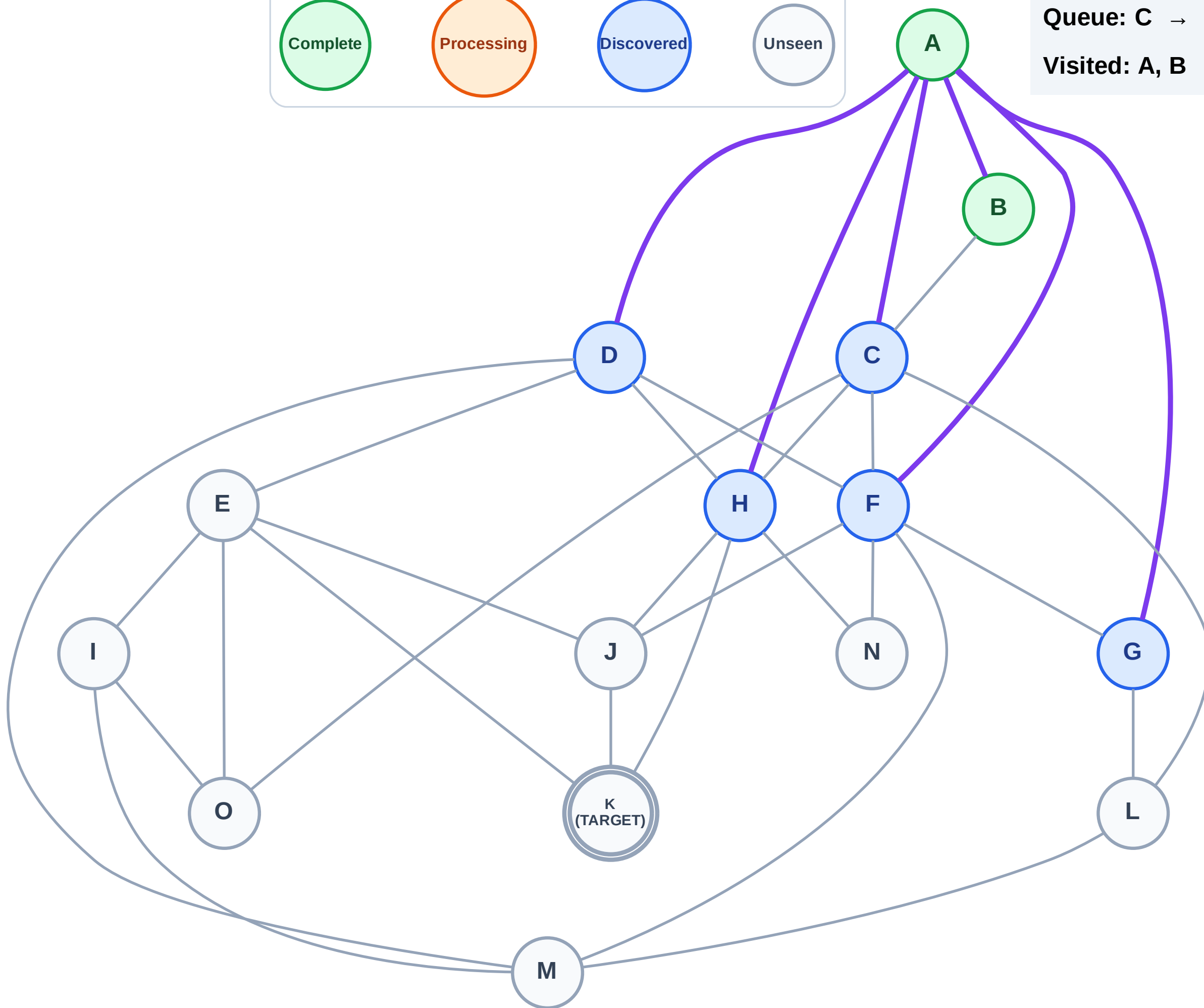
BFS Traversal

Step 5: No new neighbors from B

Legend



Queue: C → D → F → G → H
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

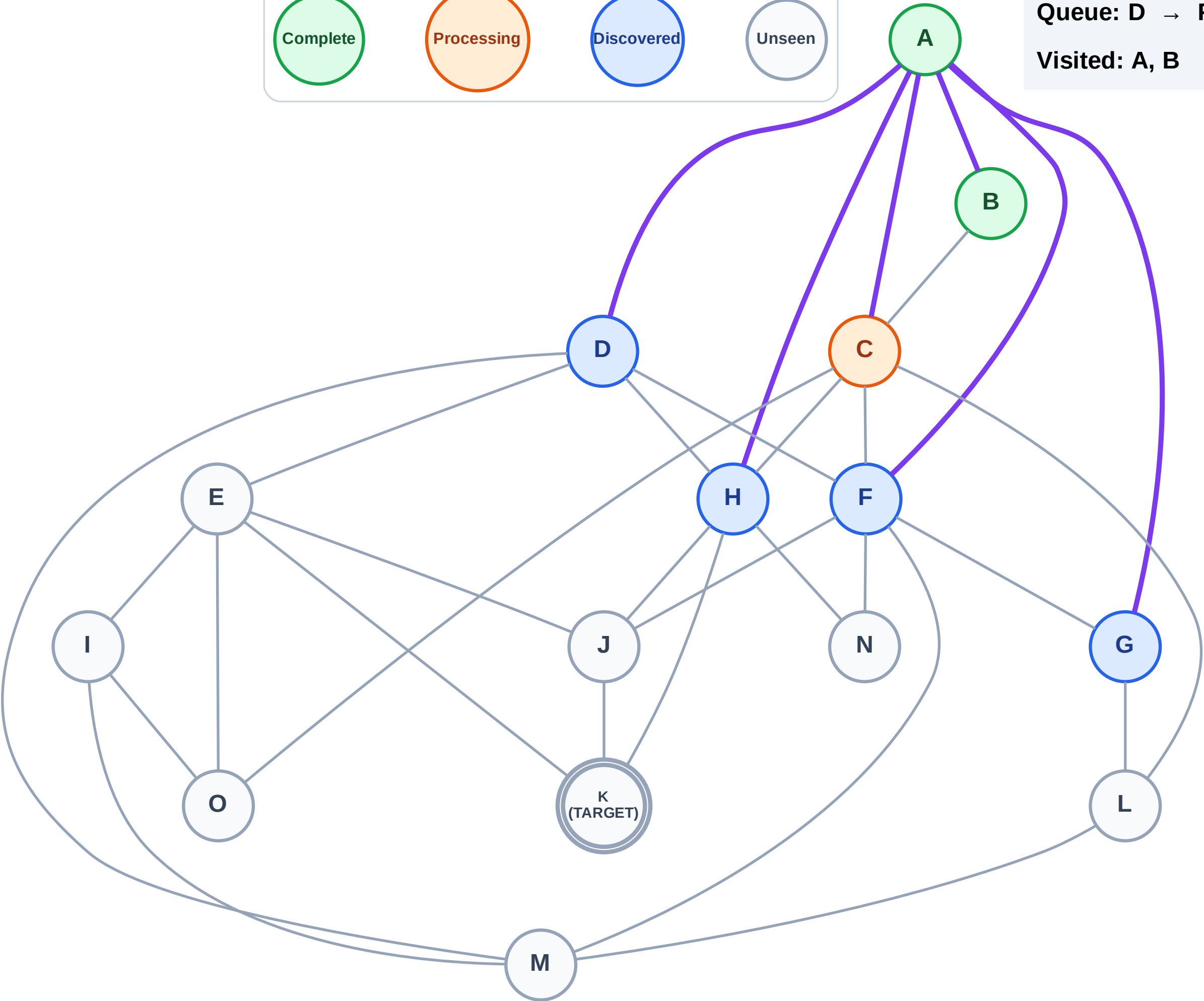
Processing

Discovered

Unseen

Queue: D → F → G → H

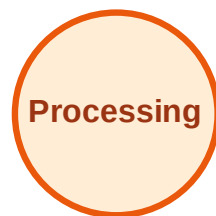
Visited: A, B



BFS Traversal

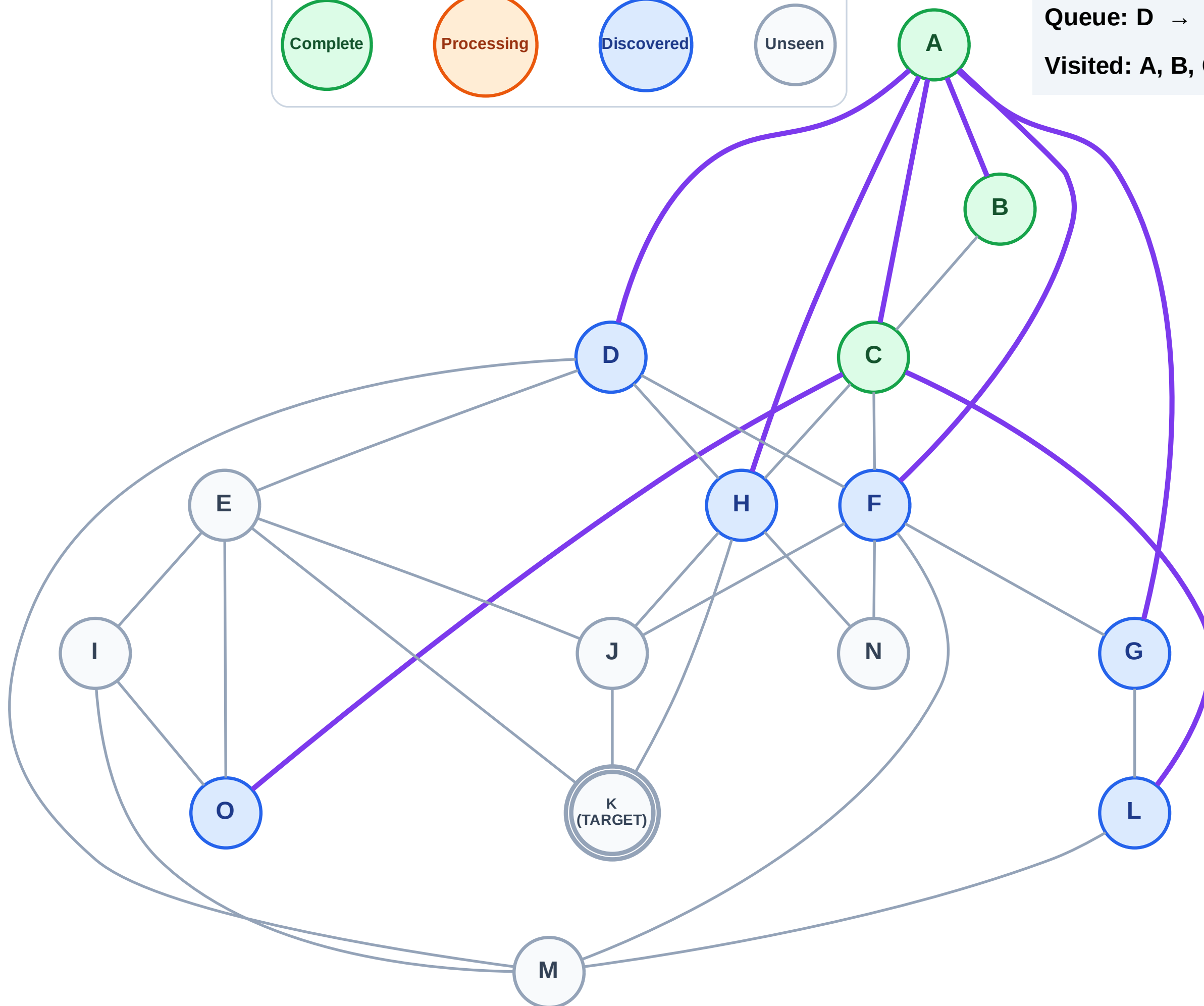
Step 7: Discover L, O from C

Legend



Queue: D → F → G → H → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

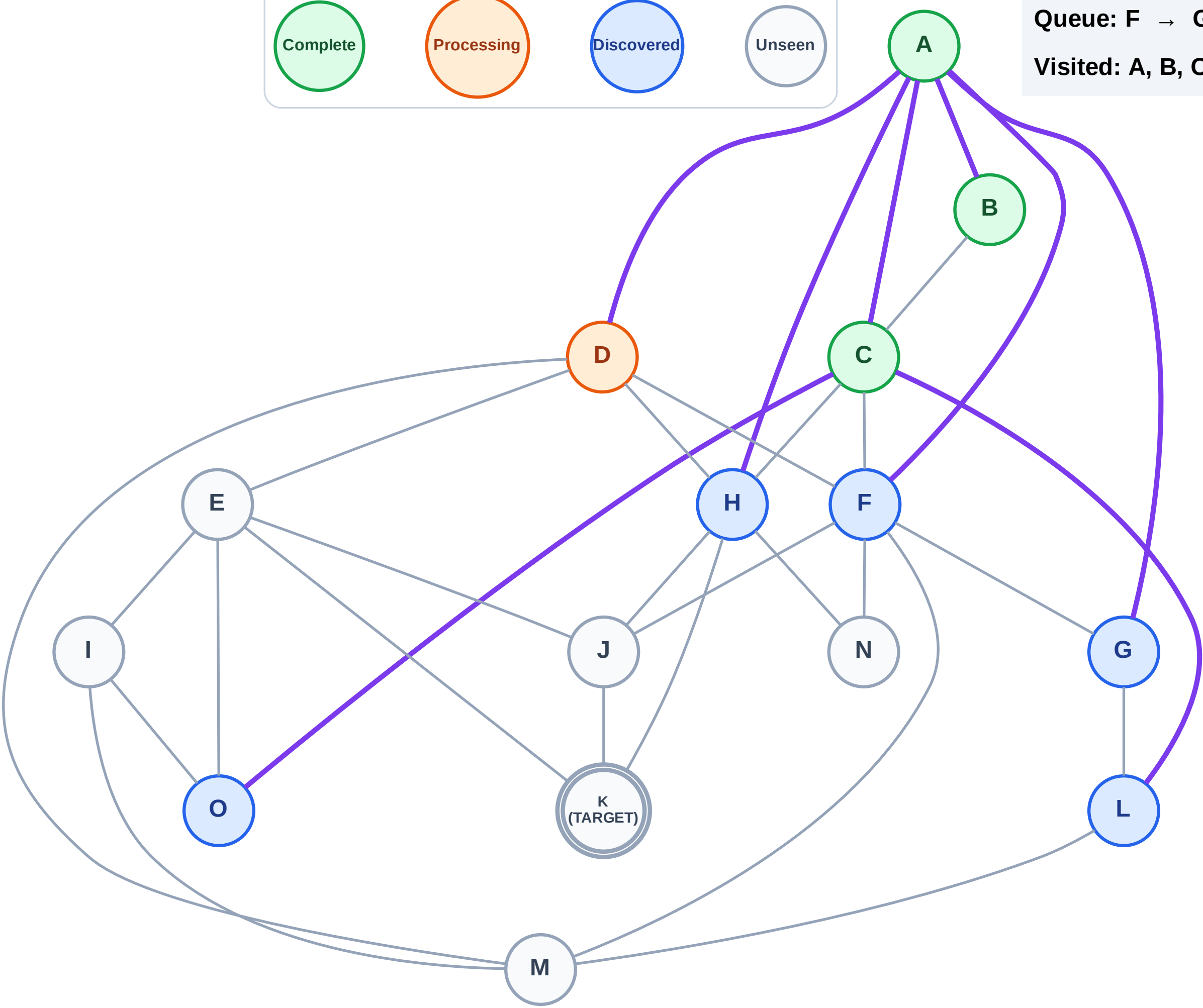
Processing

Discovered

Unseen

Queue: F → G → H → L → O

Visited: A, B, C



Step 9: Discover E, M from D

Legend



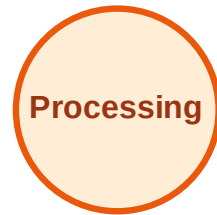
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



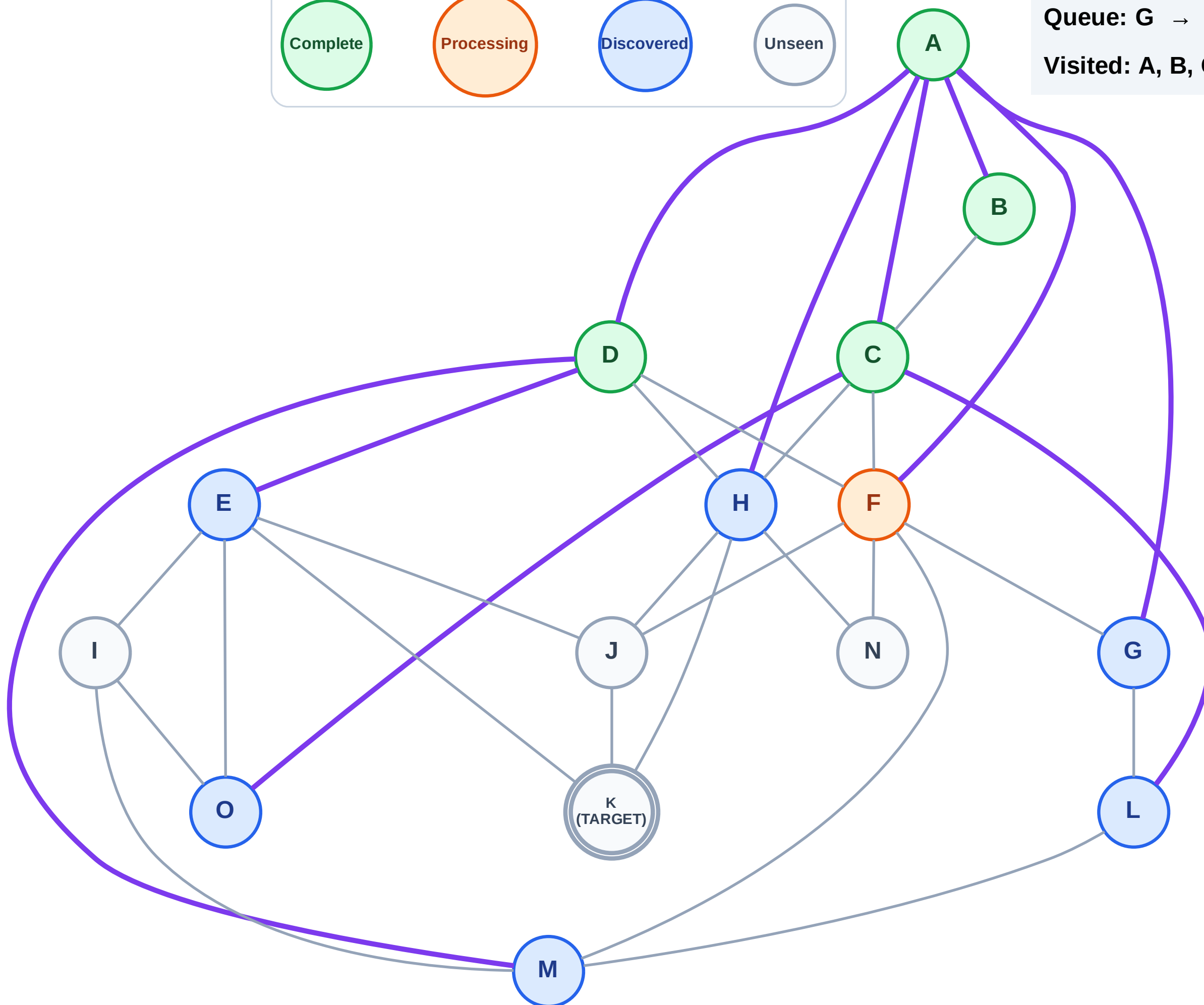
Discovered



Unseen

Queue: G → H → L → O → E → M

Visited: A, B, C, D



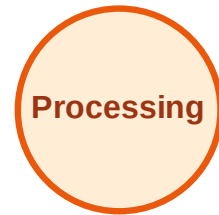
BFS Traversal

Step 11: Discover J, N from F

Legend



Complete



Processing



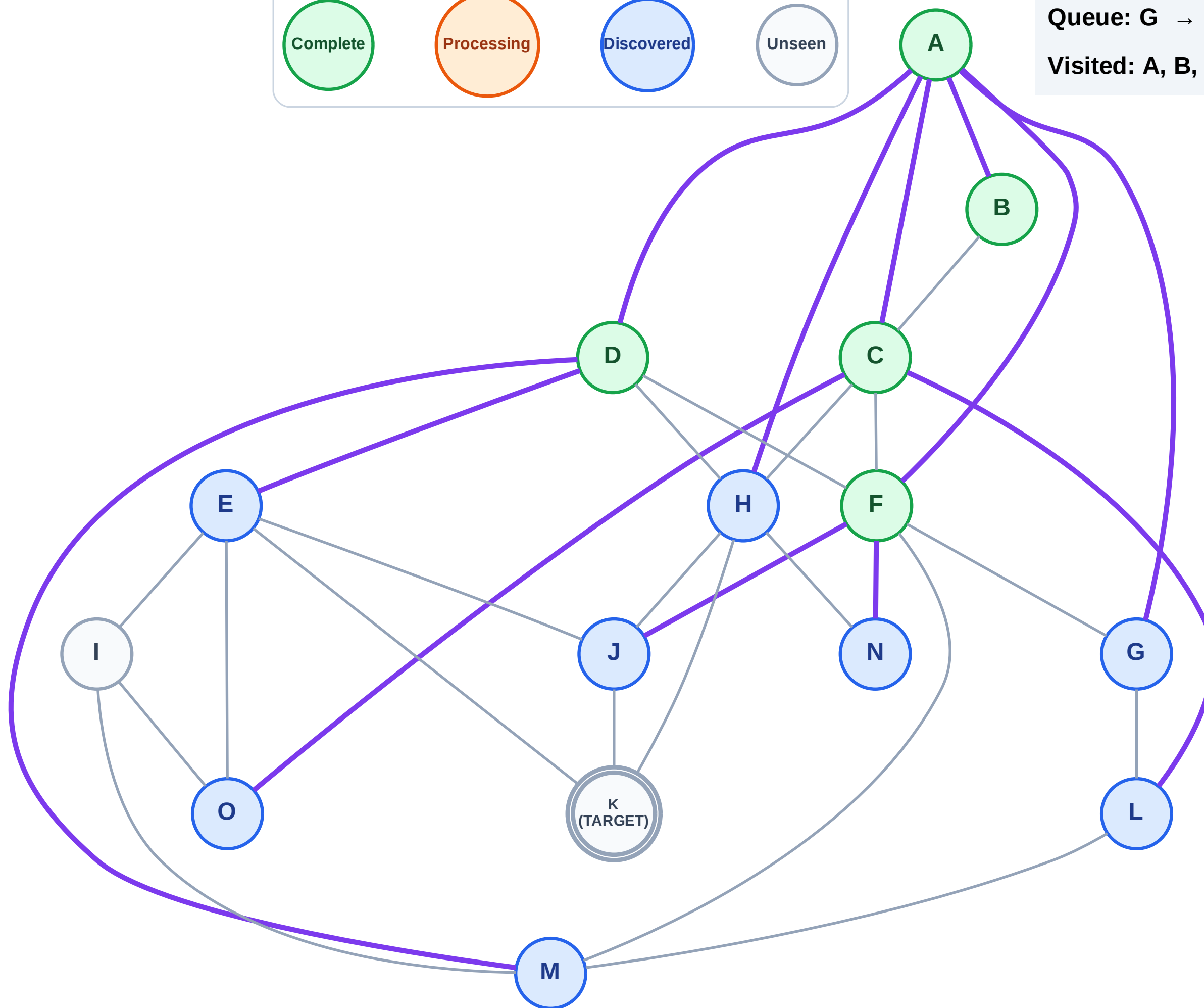
Discovered



Unseen

Queue: G → H → L → O → E → M → J → N

Visited: A, B, C, D, F



BFS Traversal

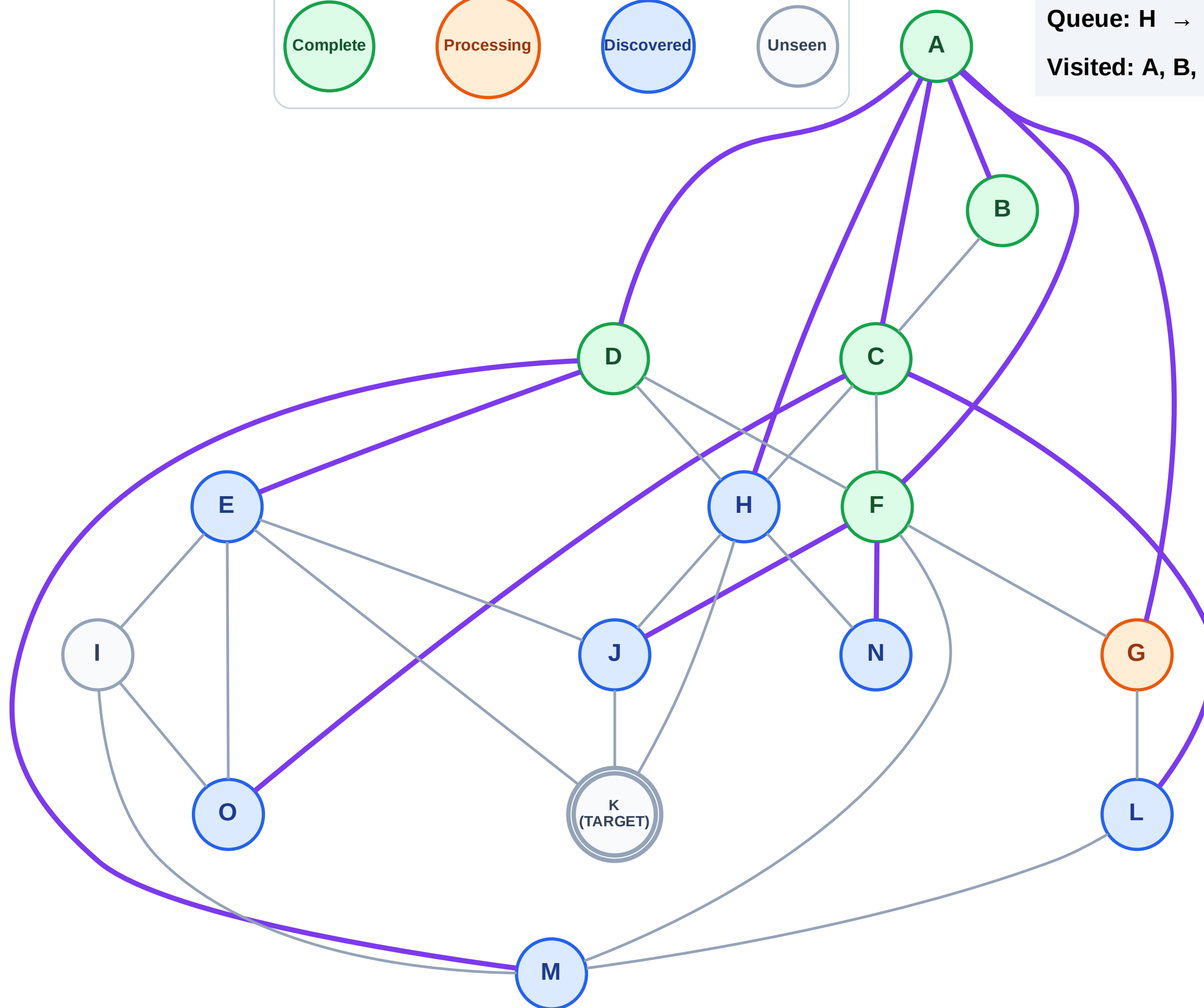
Step 12: Process node G

Legend



Queue: H → L → O → E → M → J → N

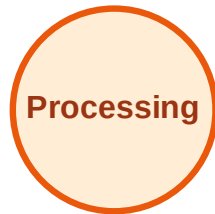
Visited: A, B, C, D, F



BFS Traversal

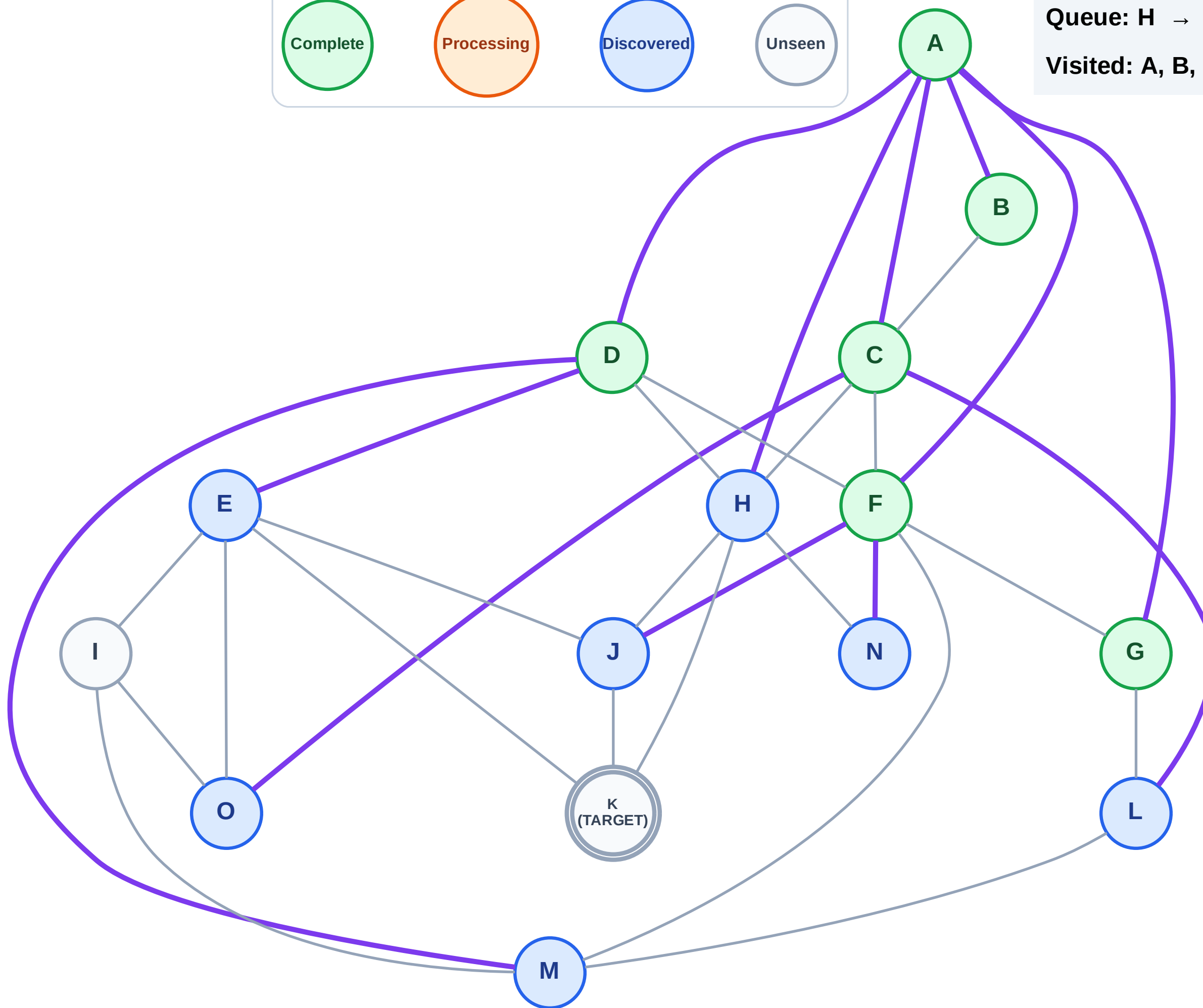
Step 13: No new neighbors from G

Legend



Queue: H → L → O → E → M → J → N

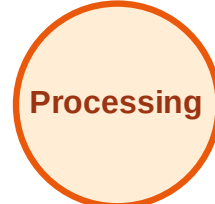
Visited: A, B, C, D, F, G



BFS Traversal

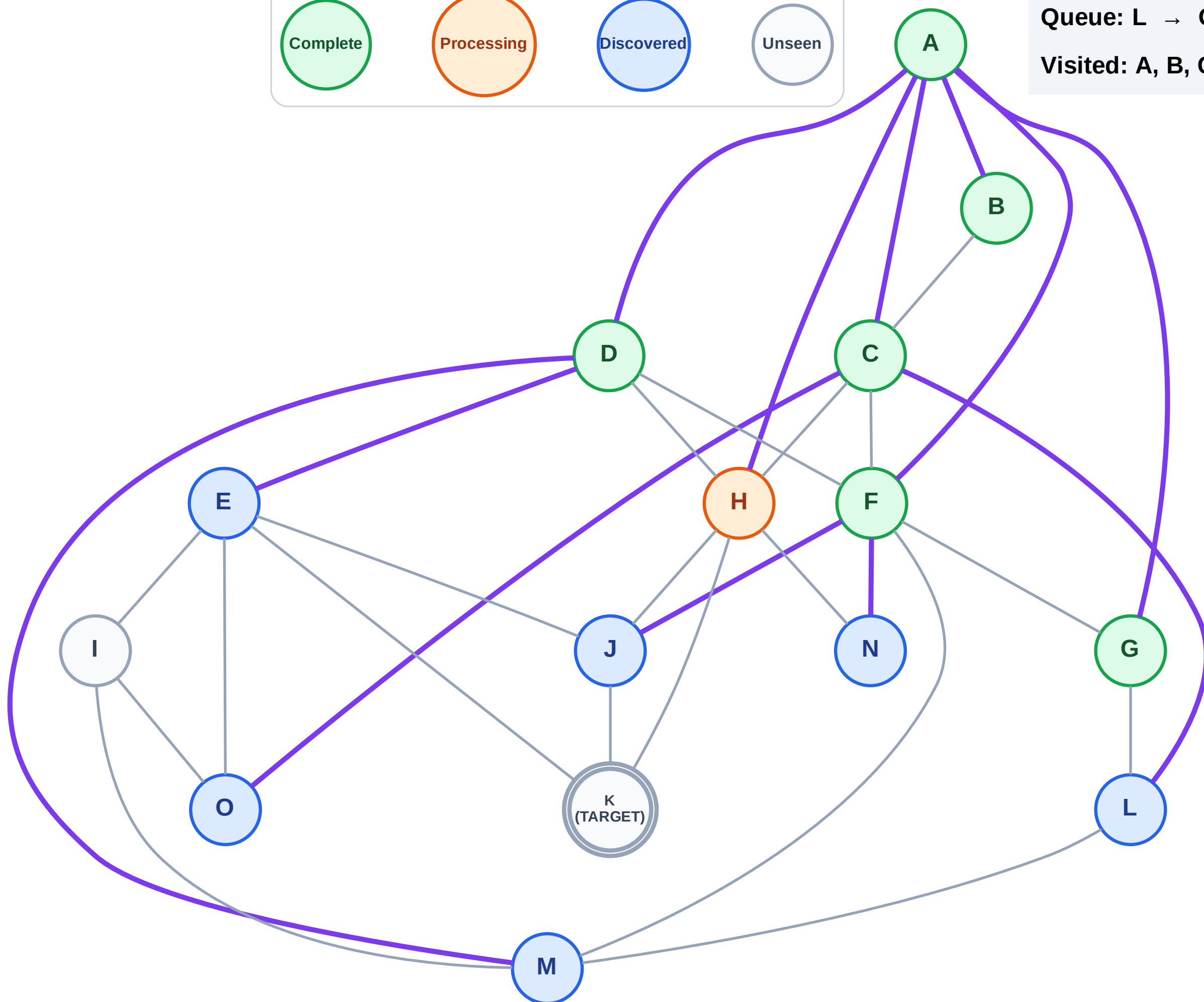
Step 14: Process node H

Legend



Queue: L → O → E → M → J → N

Visited: A, B, C, D, F, G



BFS Traversal

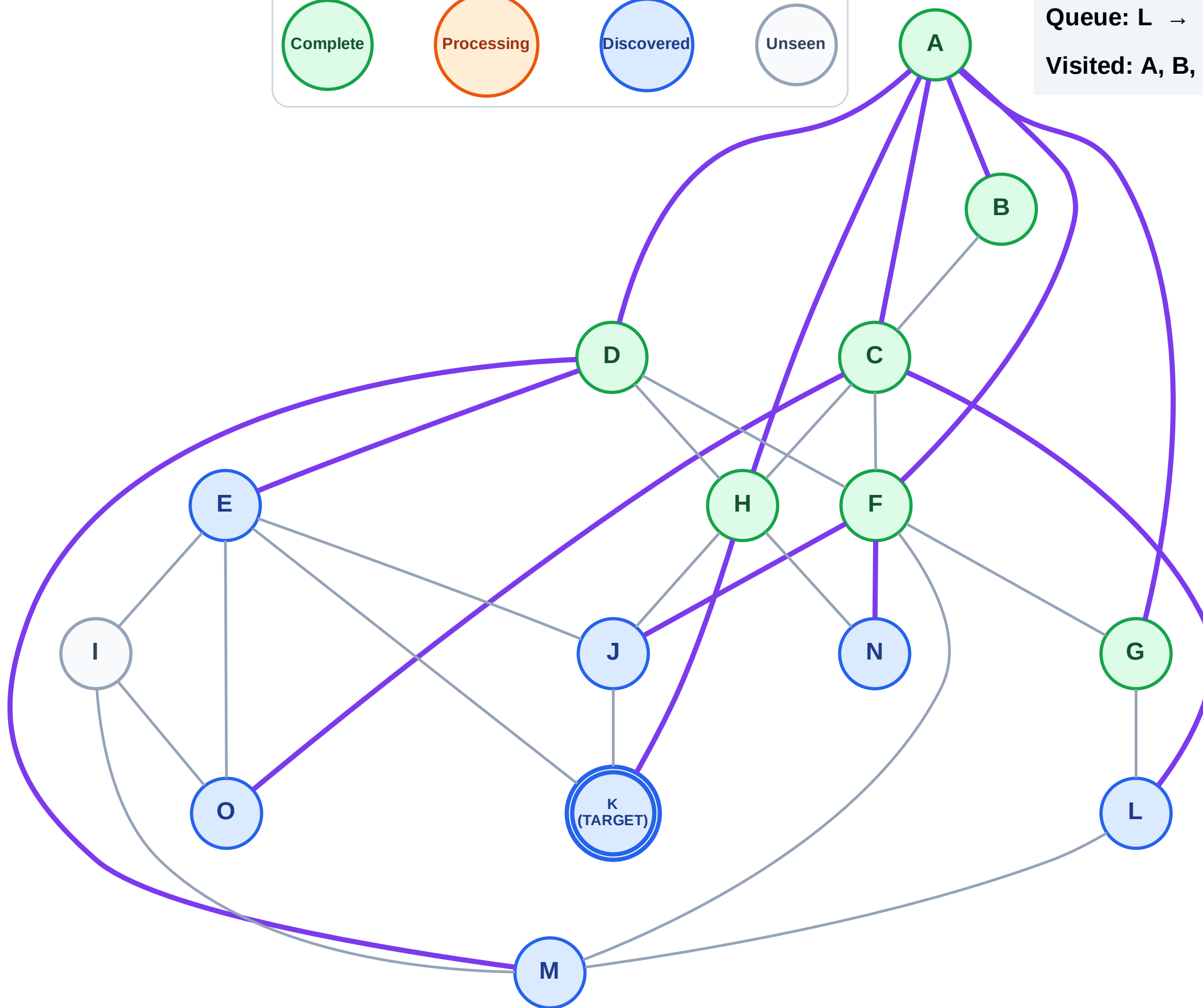
Step 15: Discover K from H

Legend



Queue: L → O → E → M → J → N → K

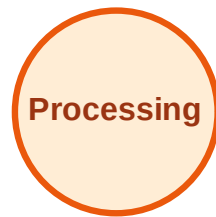
Visited: A, B, C, D, F, G, H



BFS Traversal

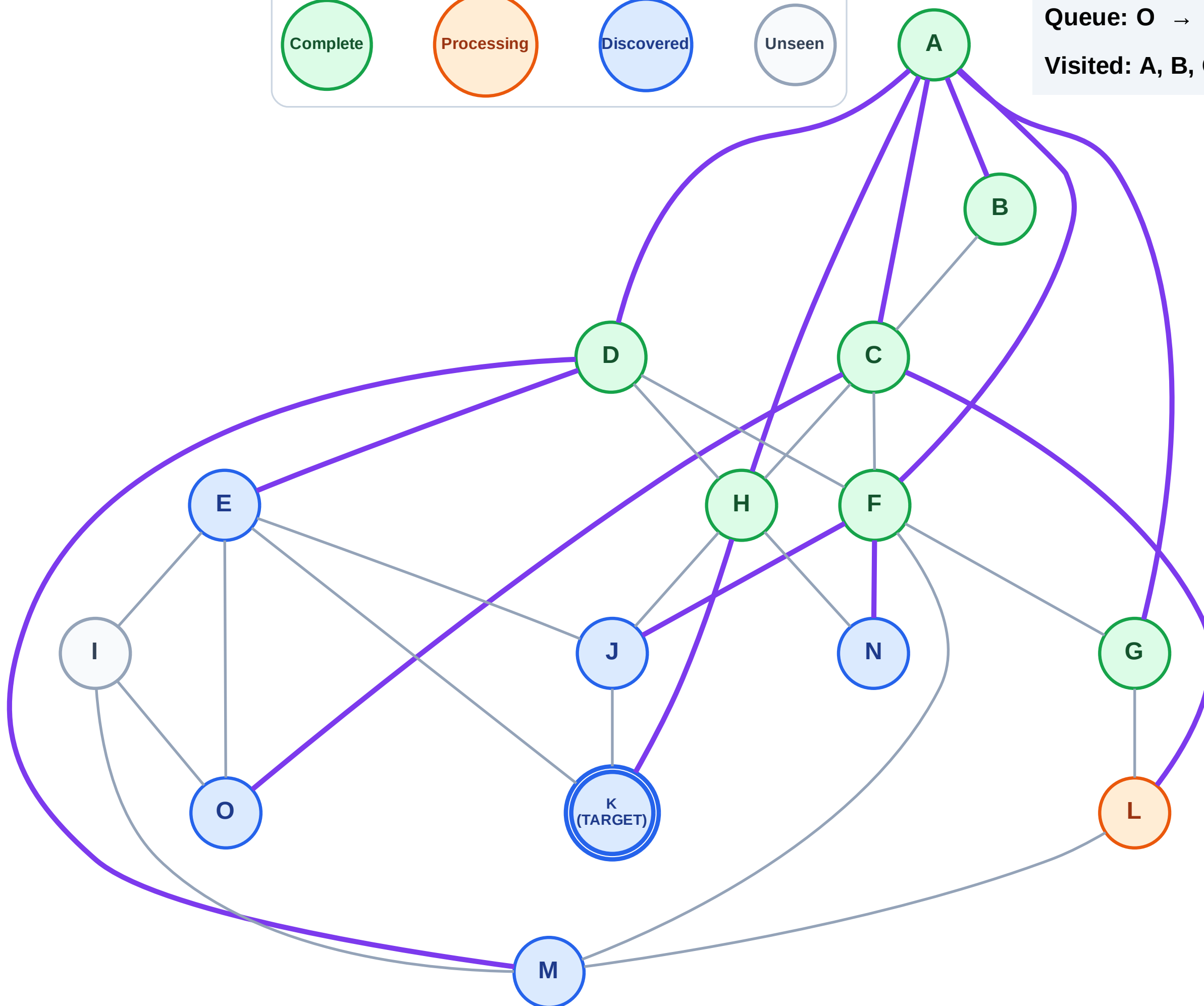
Step 16: Process node L

Legend



Queue: O → E → M → J → N → K

Visited: A, B, C, D, F, G, H



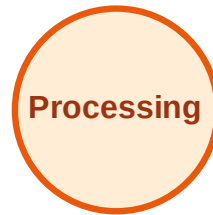
BFS Traversal

Step 17: No new neighbors from L

Legend



Complete



Processing



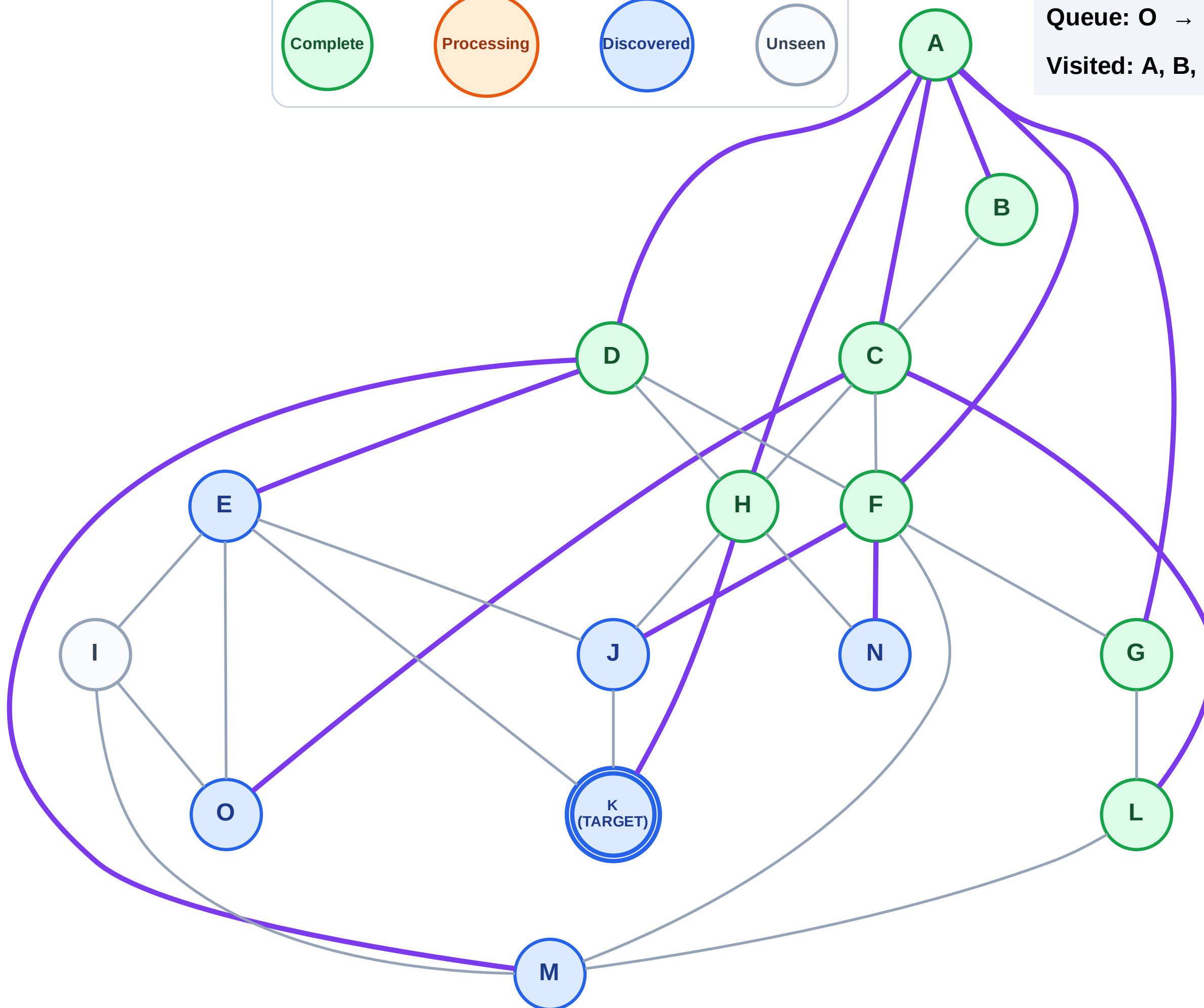
Discovered



Unseen

Queue: O → E → M → J → N → K

Visited: A, B, C, D, F, G, H, L



BFS Traversal

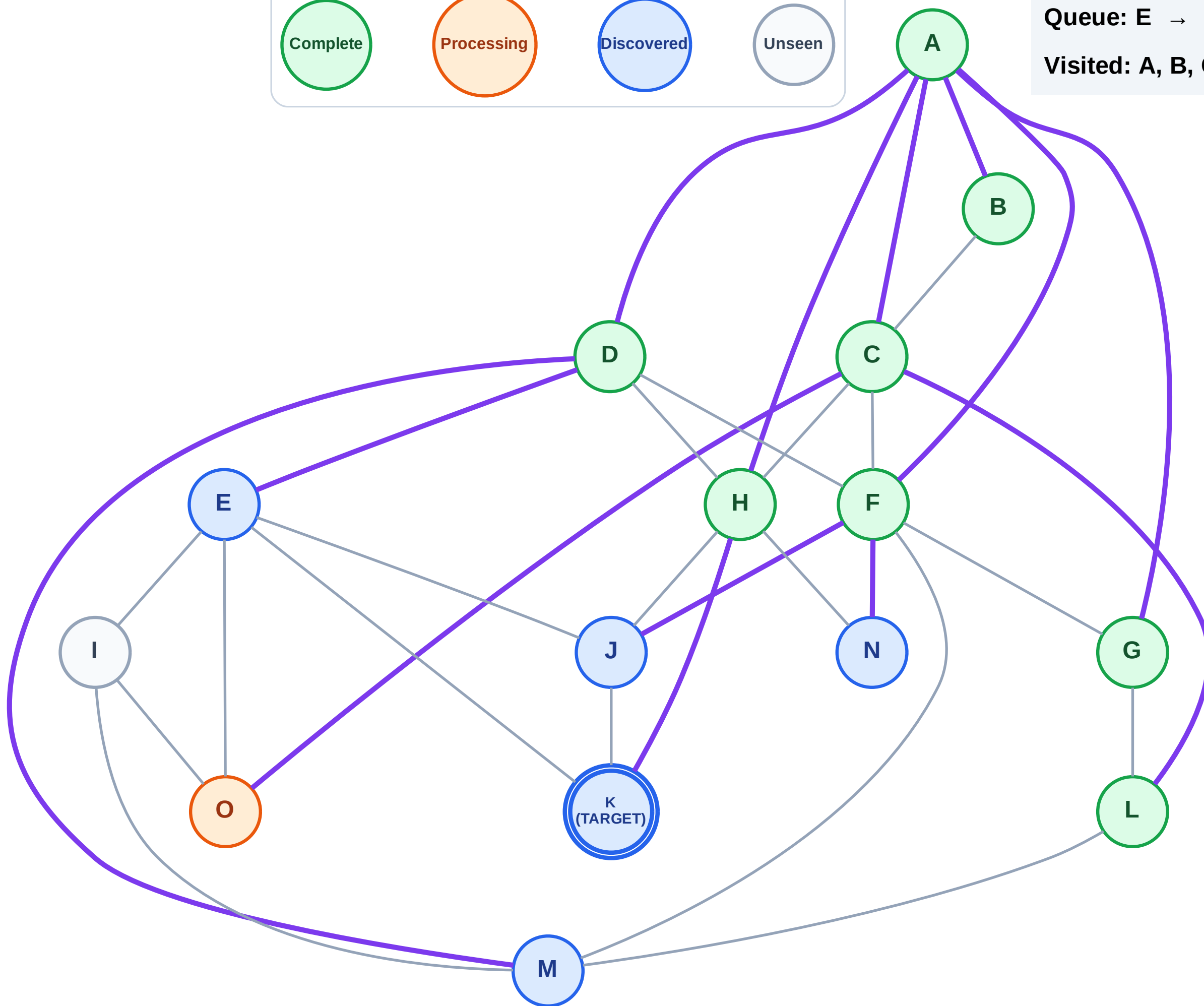
Step 18: Process node O

Legend



Queue: E → M → J → N → K

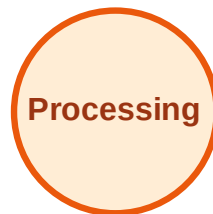
Visited: A, B, C, D, F, G, H, L



BFS Traversal

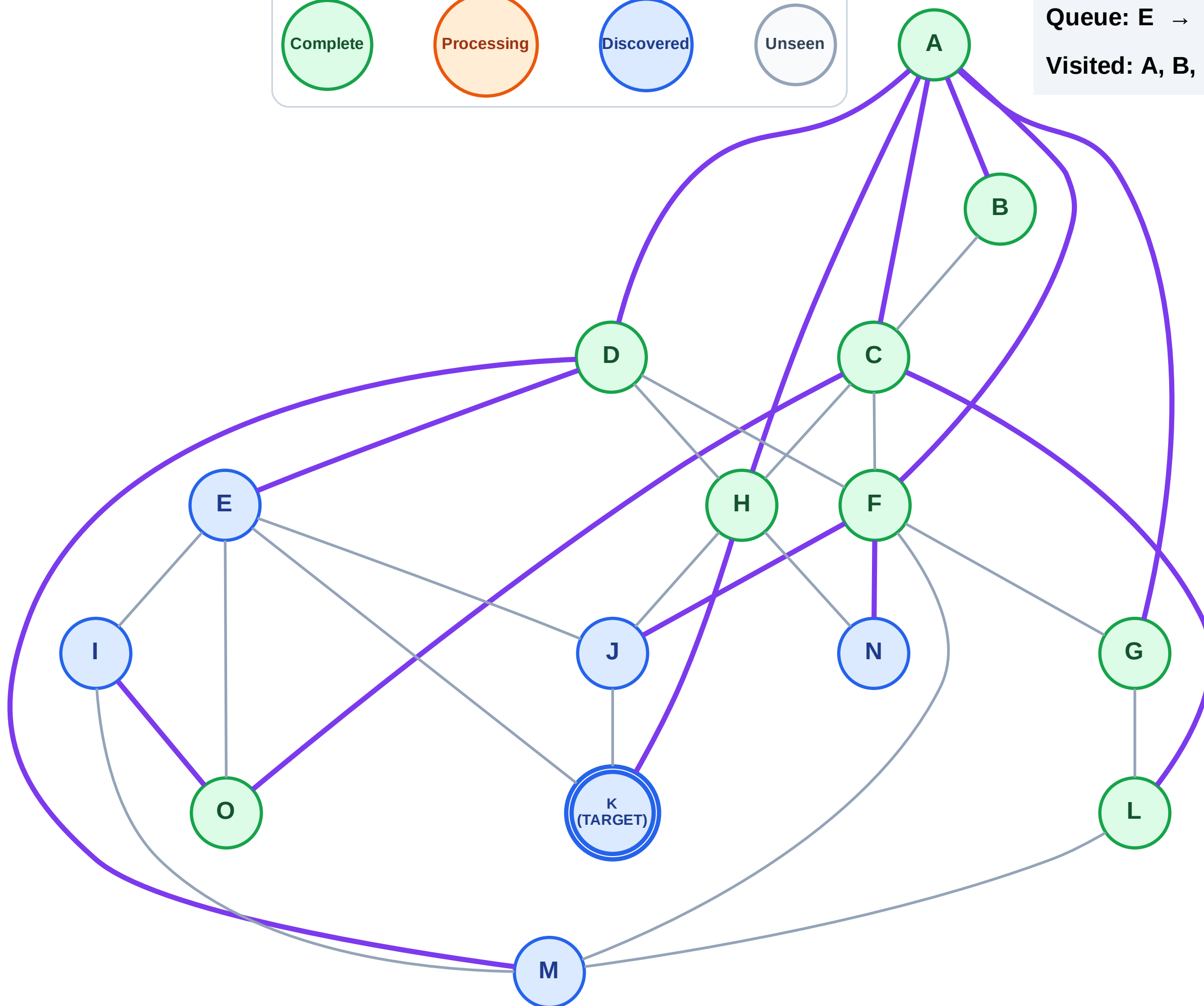
Step 19: Discover I from O

Legend



Queue: E → M → J → N → K → I

Visited: A, B, C, D, F, G, H, L, O



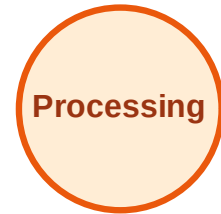
BFS Traversal

Step 20: Process node E

Legend



Complete



Processing



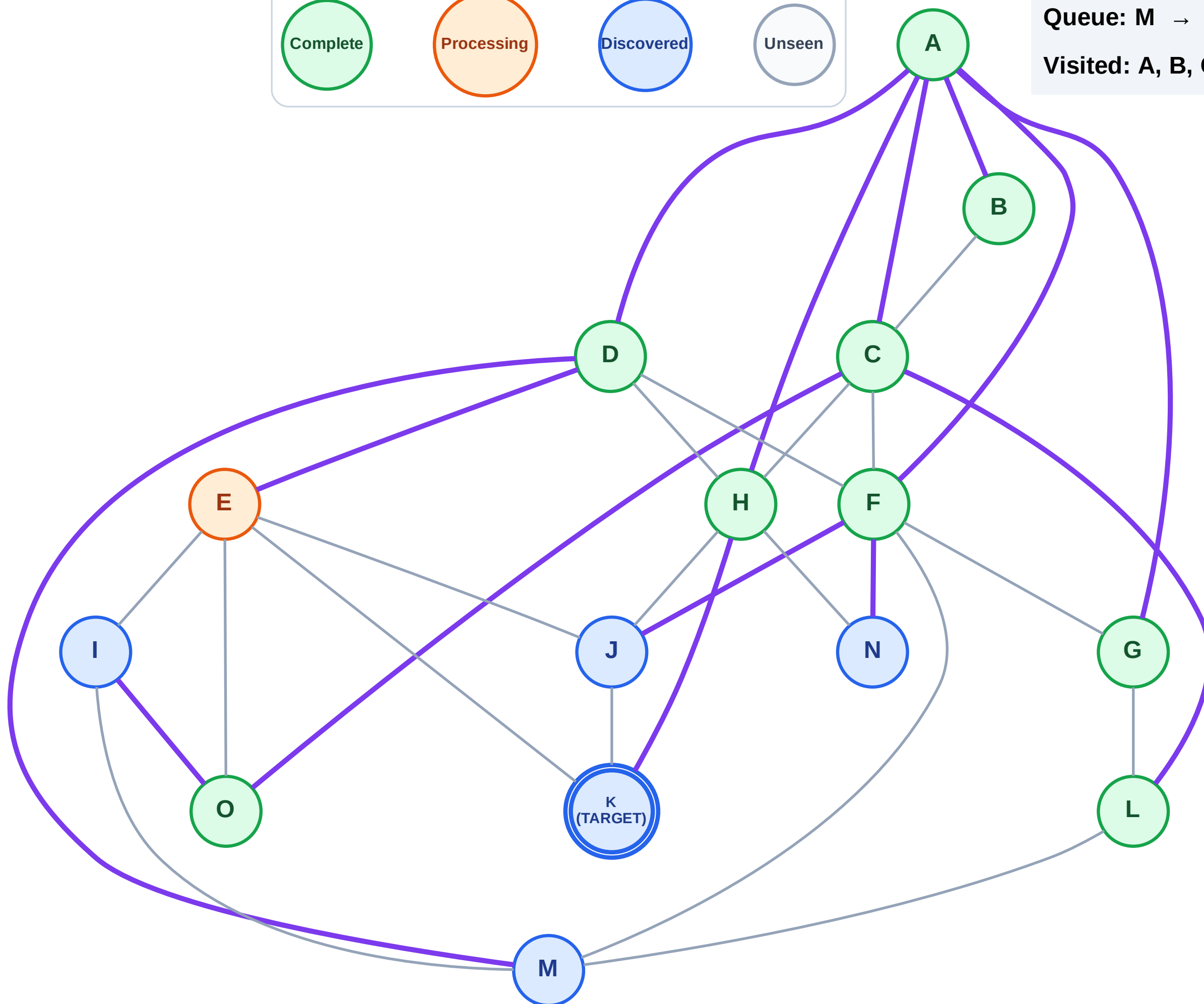
Discovered



Unseen

Queue: M → J → N → K → I

Visited: A, B, C, D, F, G, H, L, O



Step 21: No new neighbors from E

Legend



Complete Processing Discovered Unseen

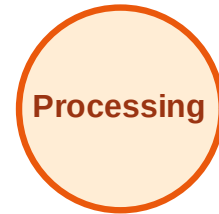
The graph shows a search process with the following nodes and edges:

- Nodes:** A, B, C, D, E, F, G, H, I, J, K (TARGET), L, M, N, O.
- Colors:**
 - Complete (Green):** A, B, C, D, E, F, G, H, L, O.
 - Processing (Orange):** None.
 - Discovered (Blue):** I, J, K (TARGET), M, N.
 - Unseen (Grey):** None.
- Queue:** M →
- Visited:** A, B, C
- Edges:**
 - A is connected to B, C, D, E, F, G, H, I, J, K, L, M, N, O.
 - B is connected to C.
 - C is connected to D, E, F, G, H, I, J, K, L, M, N, O.
 - D is connected to E, F, G, H, I, J, K, L, M, N, O.
 - E is connected to F, G, H, I, J, K, L, M, N, O.
 - F is connected to G, H, I, J, K, L, M, N, O.
 - G is connected to H, I, J, K, L, M, N, O.
 - H is connected to I, J, K, L, M, N, O.
 - I is connected to J, K, L, M, N, O.
 - J is connected to K, L, M, N, O.
 - K (TARGET) is connected to L, M, N, O.
 - L is connected to M, N, O.
 - M is connected to N, O.
 - N is connected to O.
 - O is connected to K (TARGET).

BFS Traversal

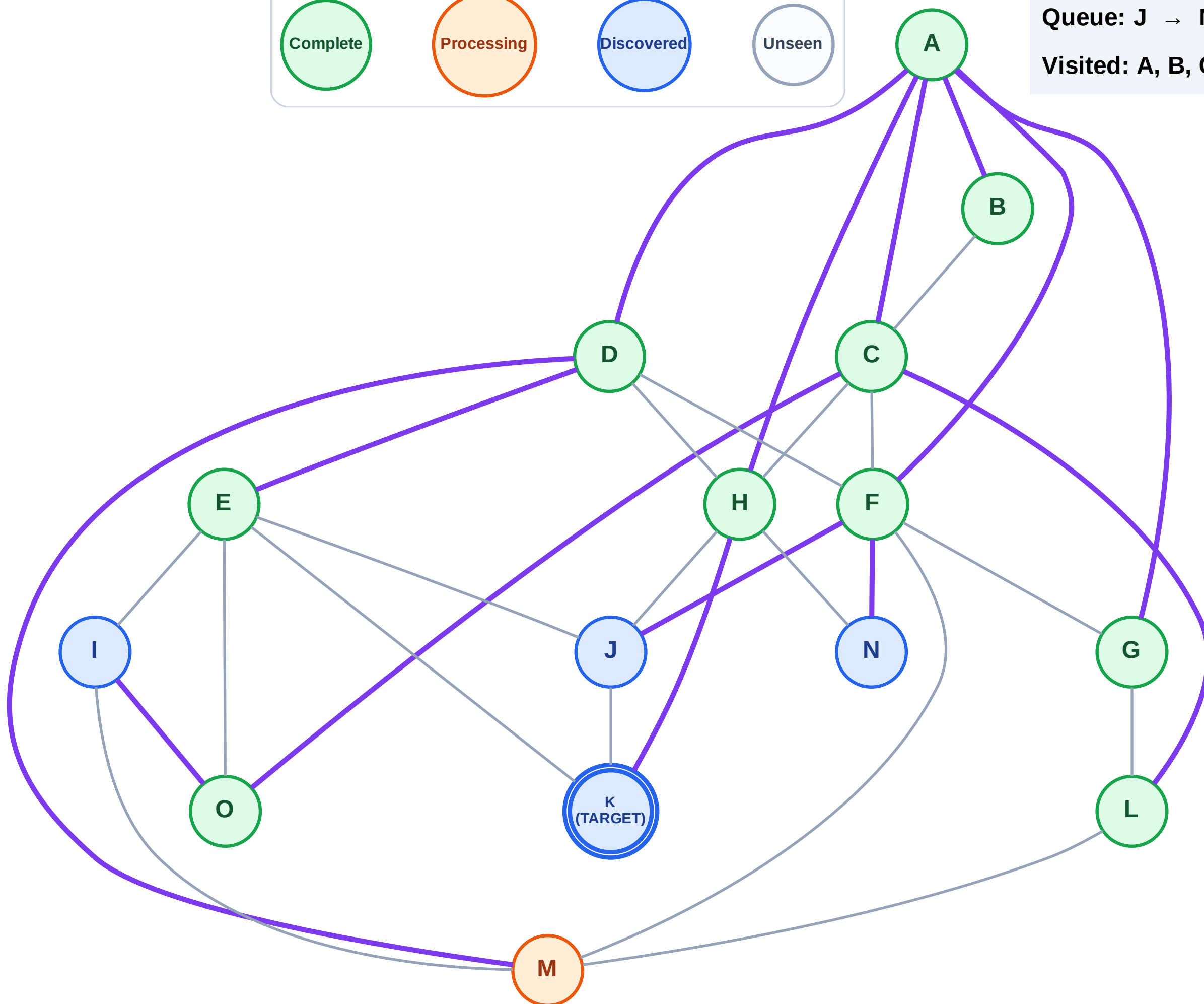
Step 22: Process node M

Legend



Queue: J → N → K → I

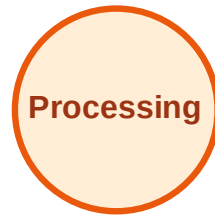
Visited: A, B, C, D, E, F, G, H, L, O



BFS Traversal

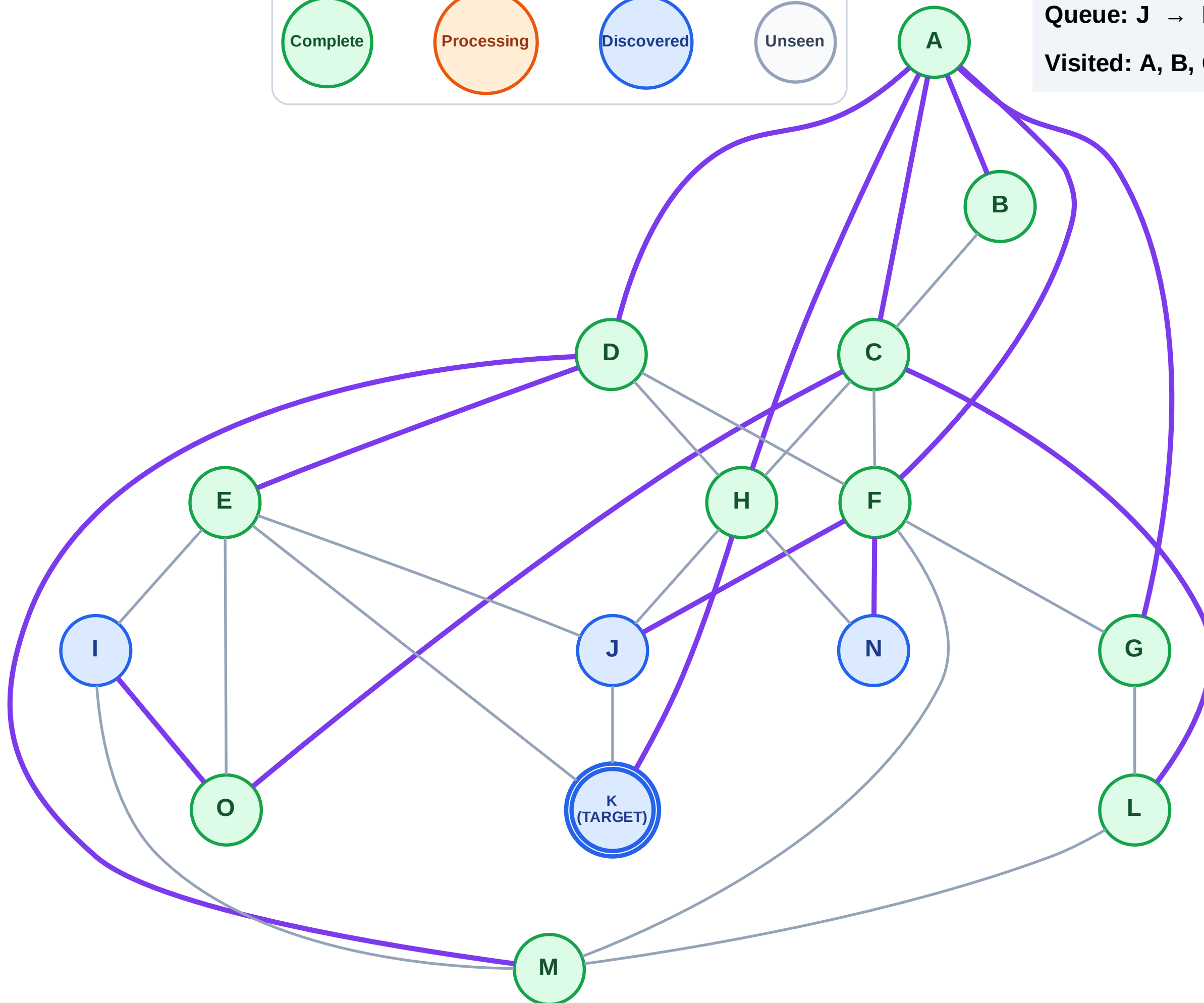
Step 23: No new neighbors from M

Legend



Queue: J → N → K → I

Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

Step 24: Process node J

Legend

Complete

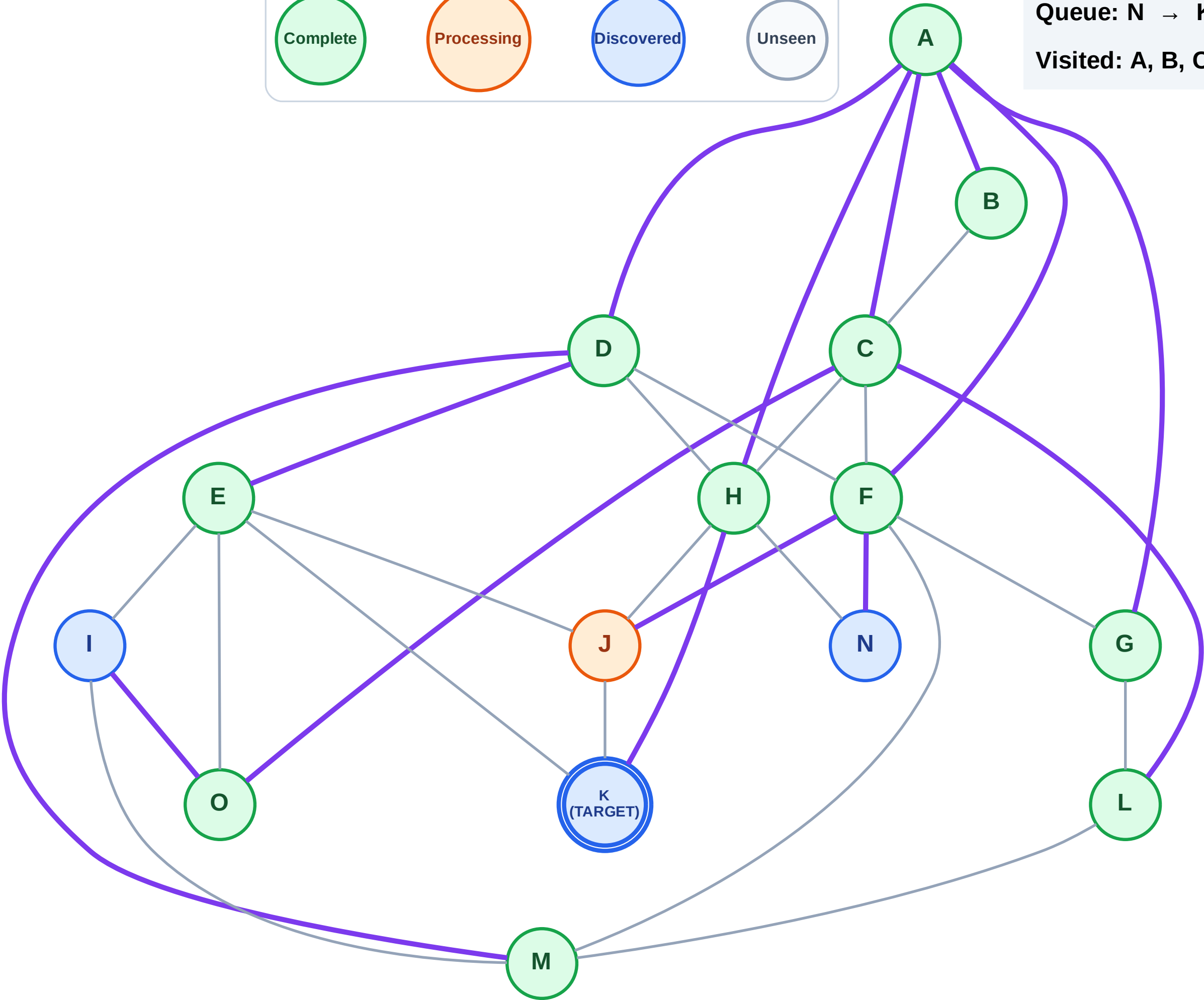
Processing

Discovered

Unseen

Queue: N → K → I

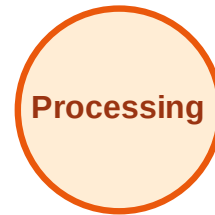
Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

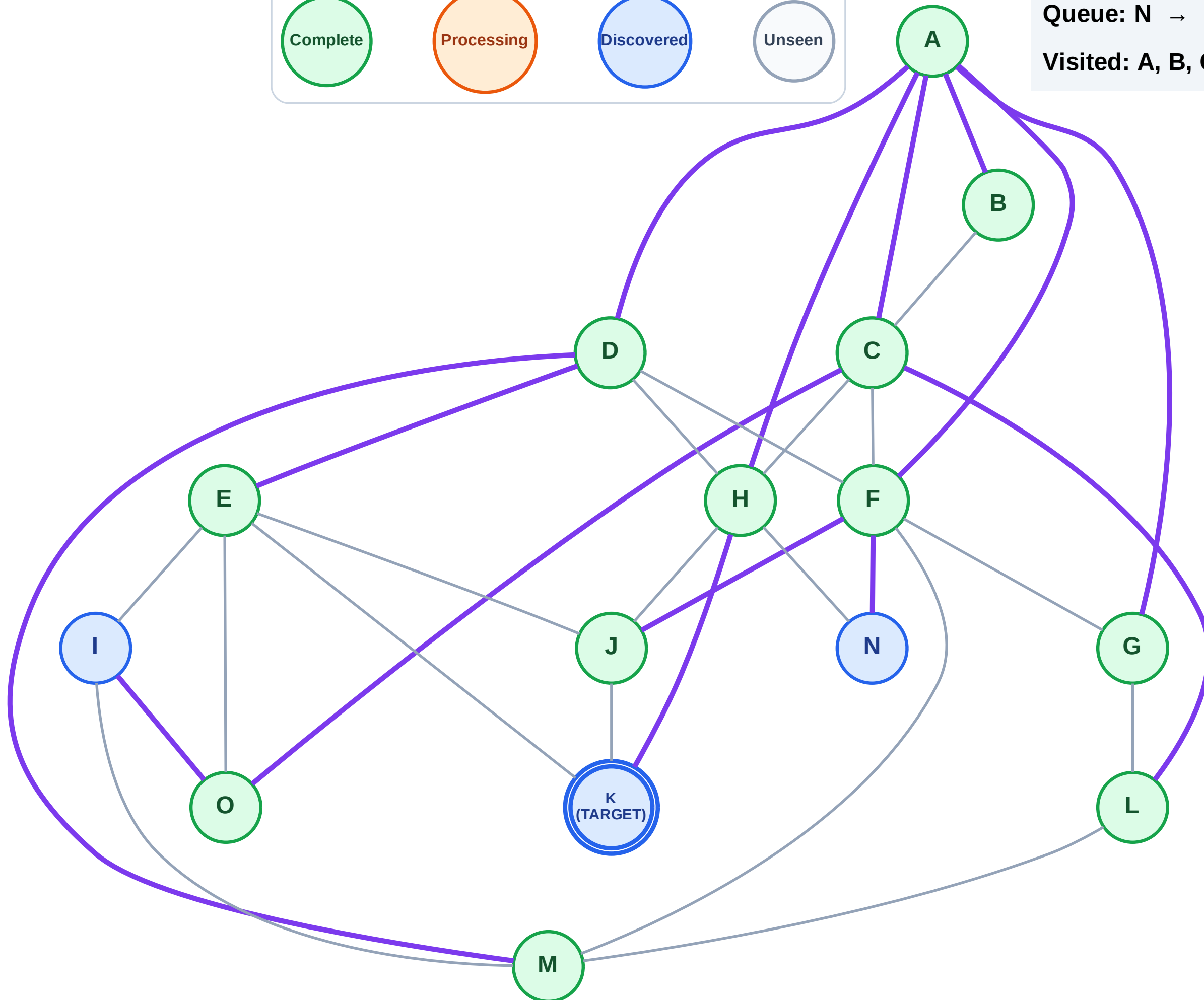
Step 25: No new neighbors from J

Legend



Queue: N → K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

Step 26: Process node N

Legend

Complete

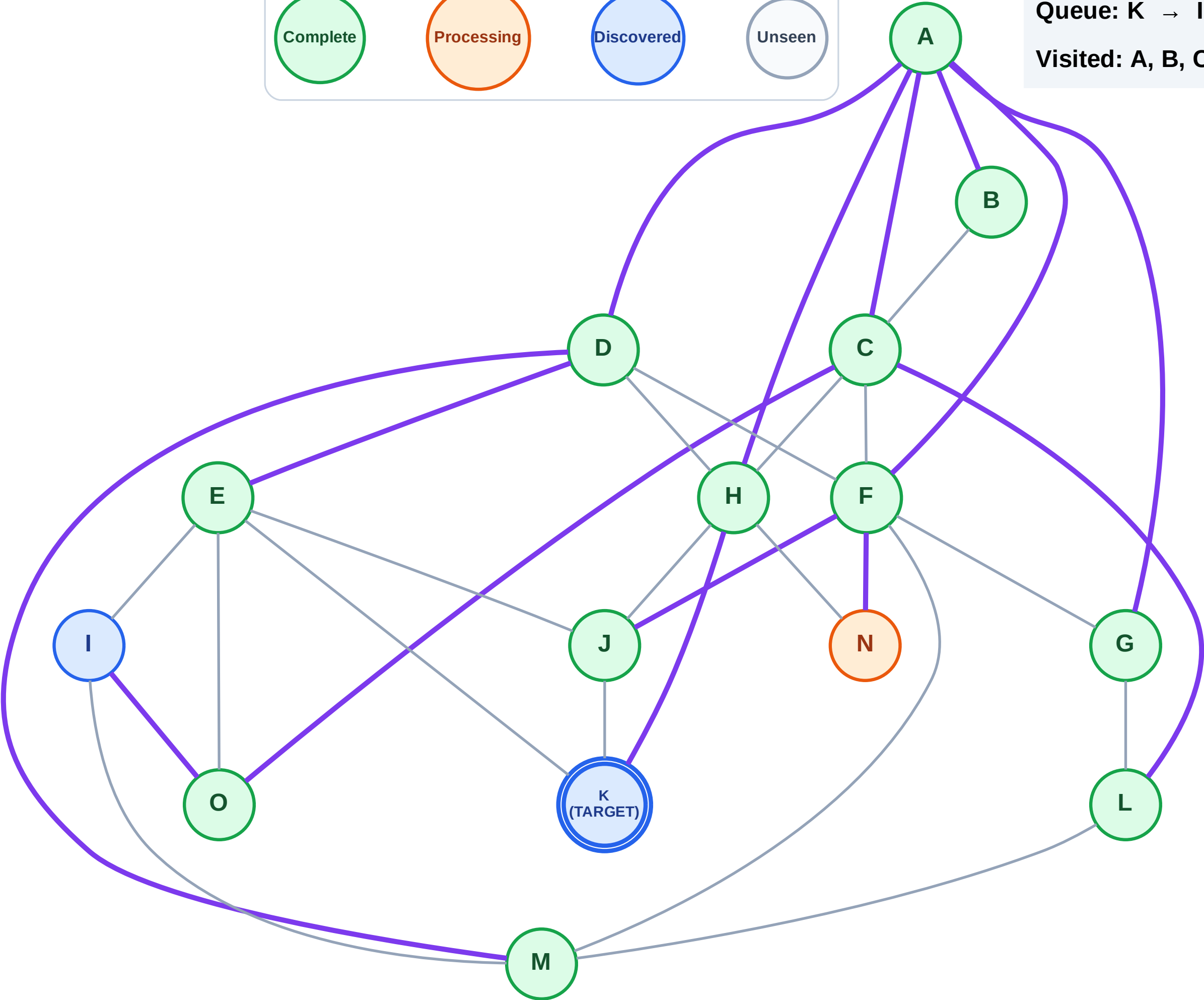
Processing

Discovered

Unseen

Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, O



BFS Traversal

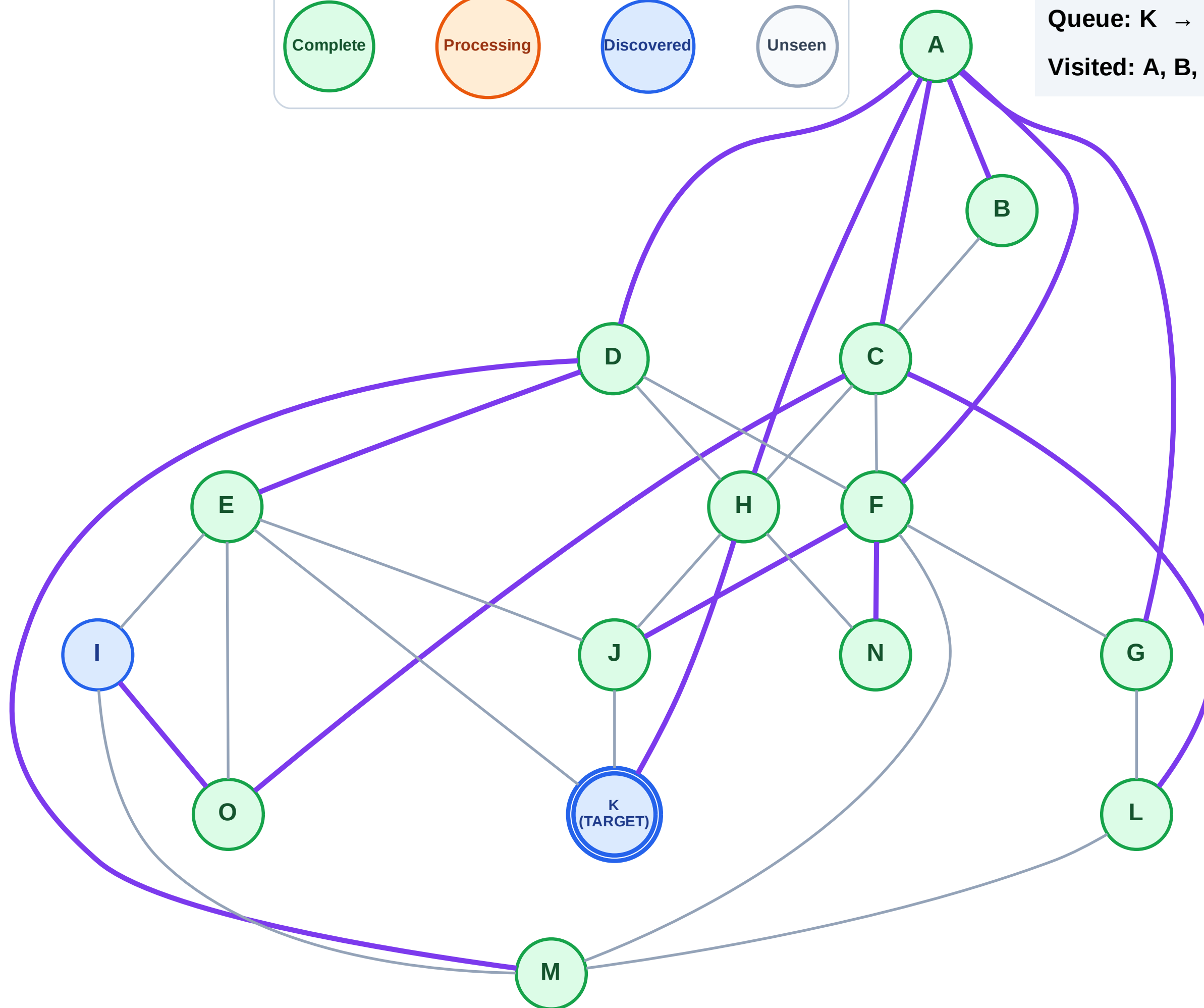
Step 27: No new neighbors from N

Legend



Queue: K → I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

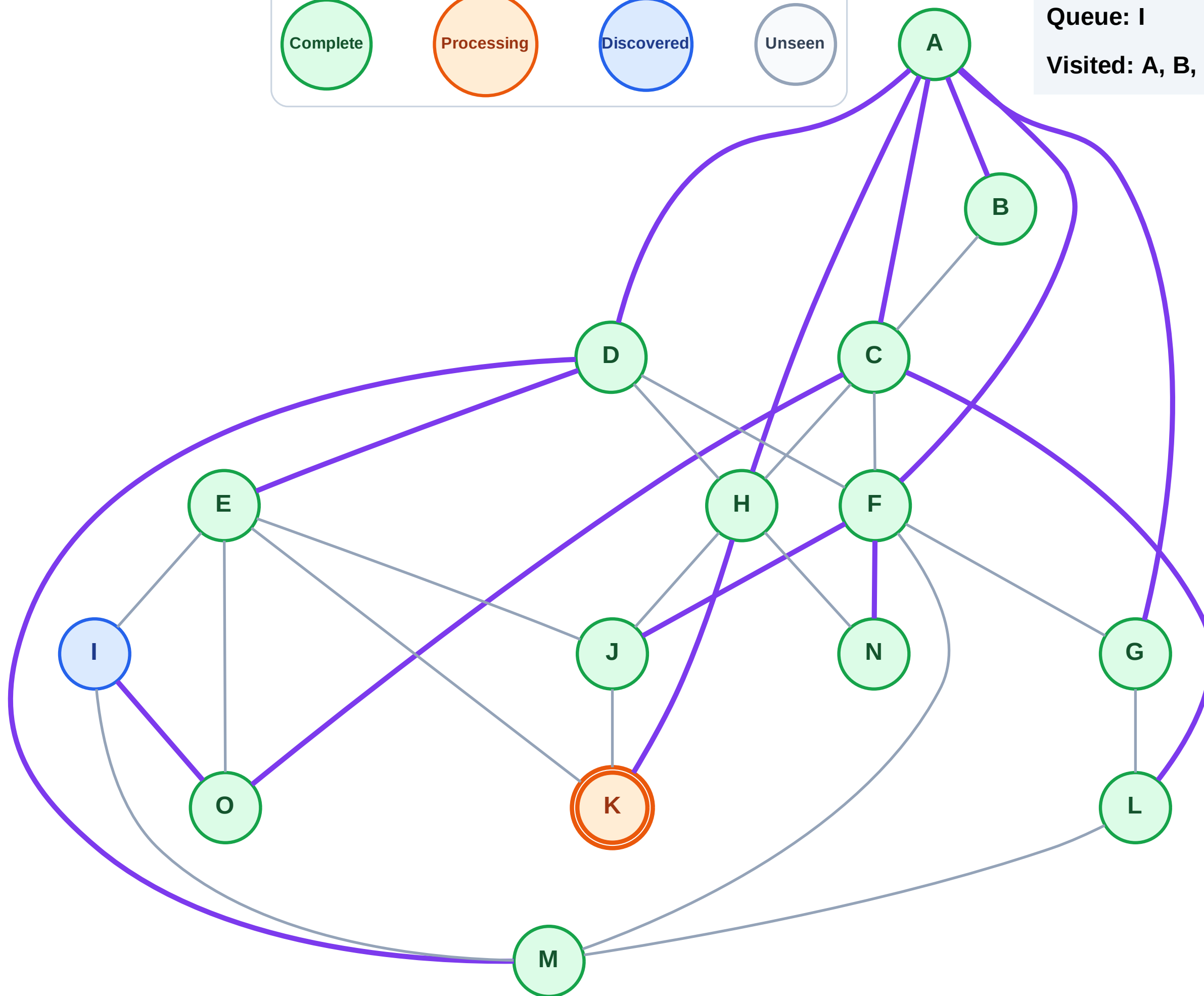
Step 28: Process node K

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



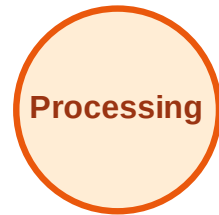
BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → H → K

Legend



Complete



Processing



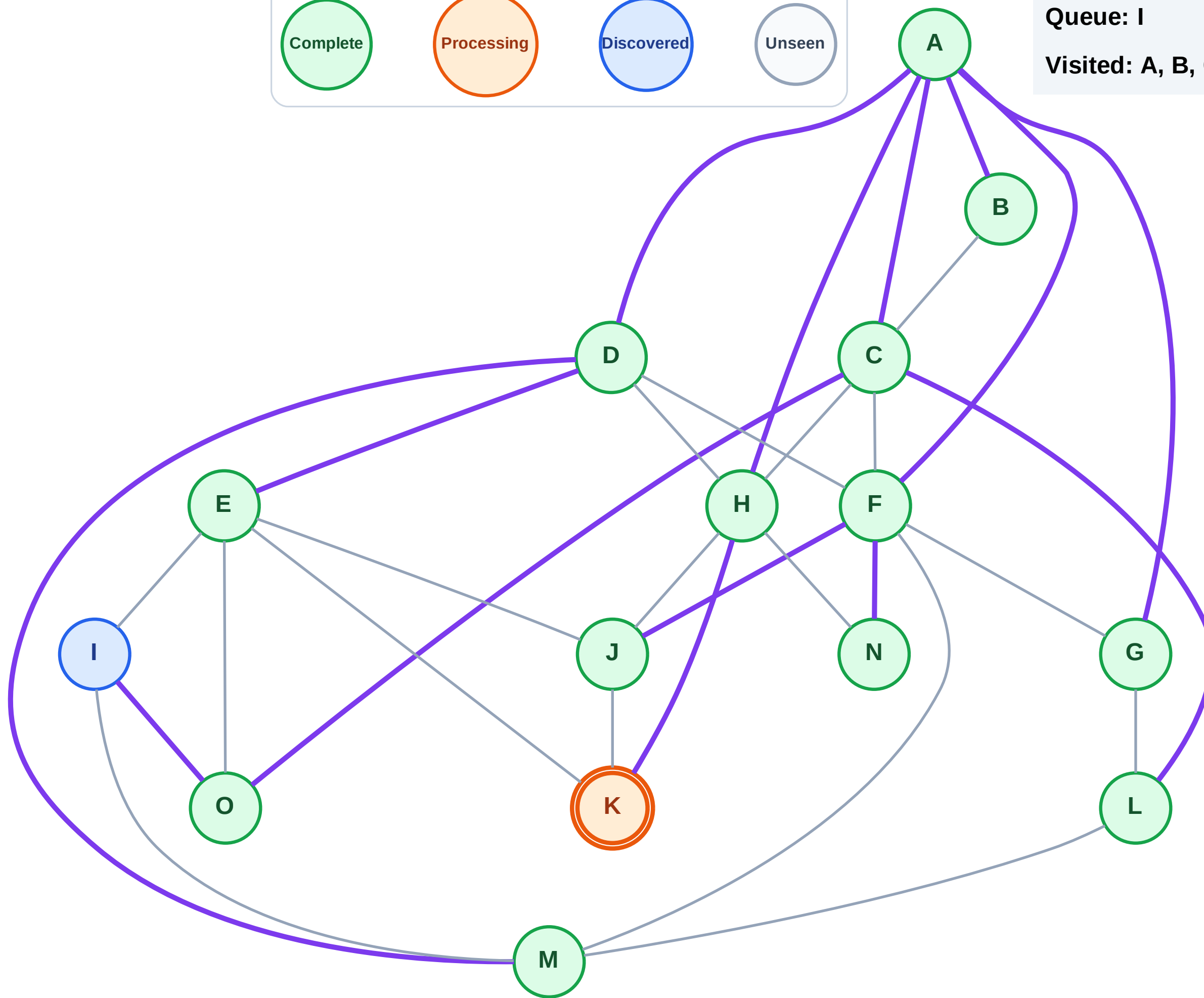
Discovered



Unseen

Queue: I

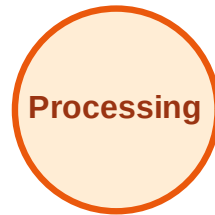
Visited: A, B, C, D, E, F, G, H, J, K, L, M, N, O



DFS Traversal

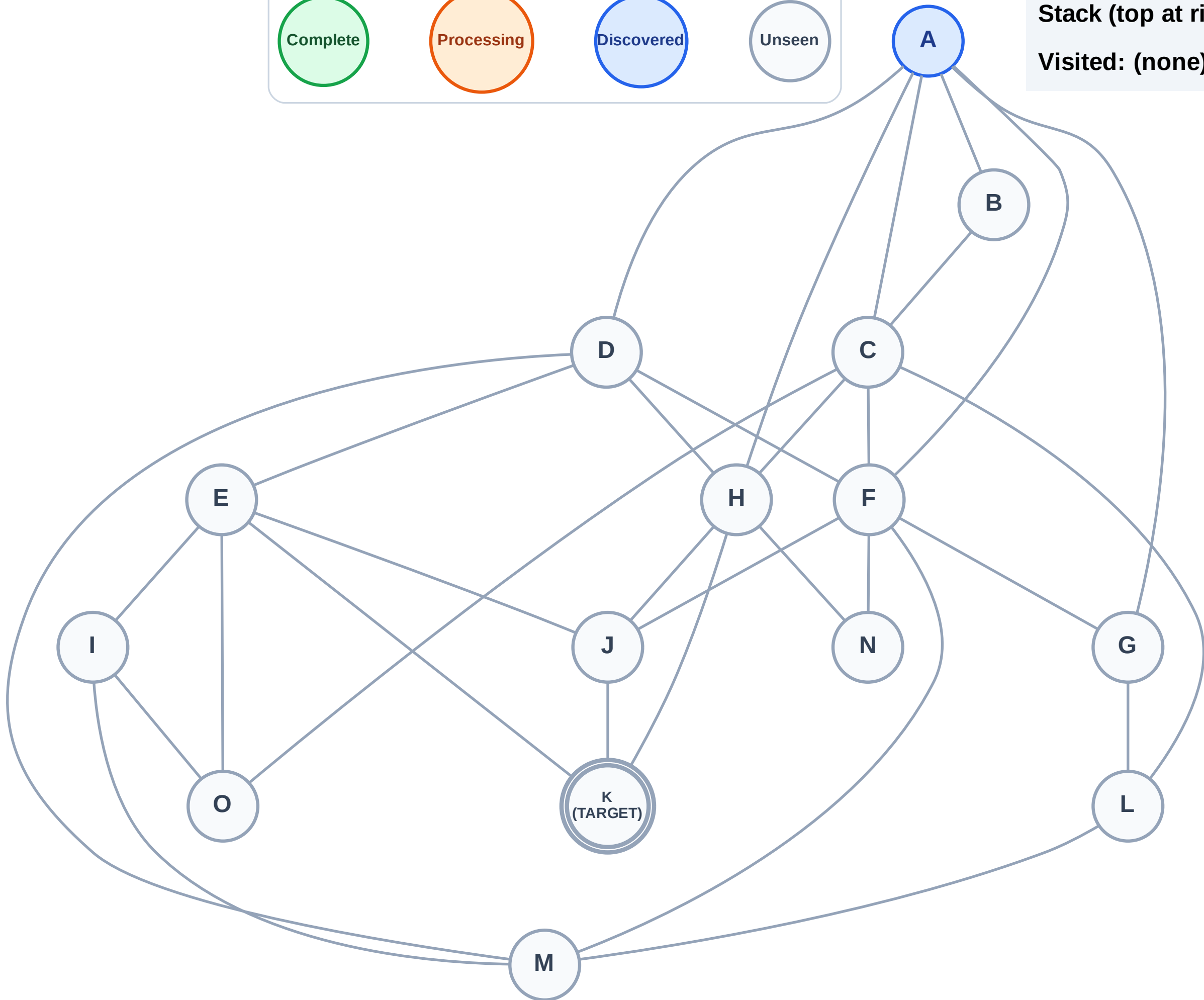
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

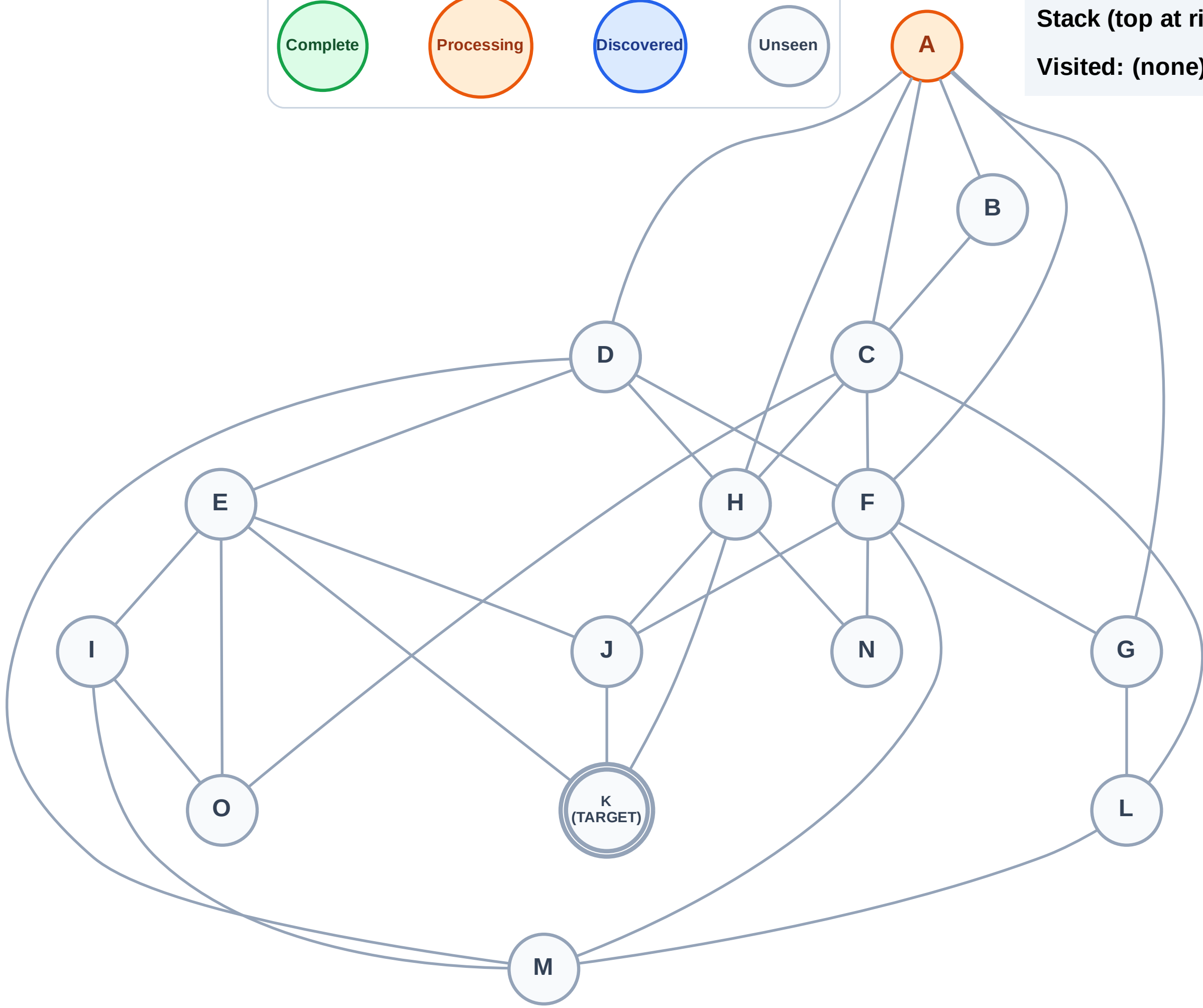
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

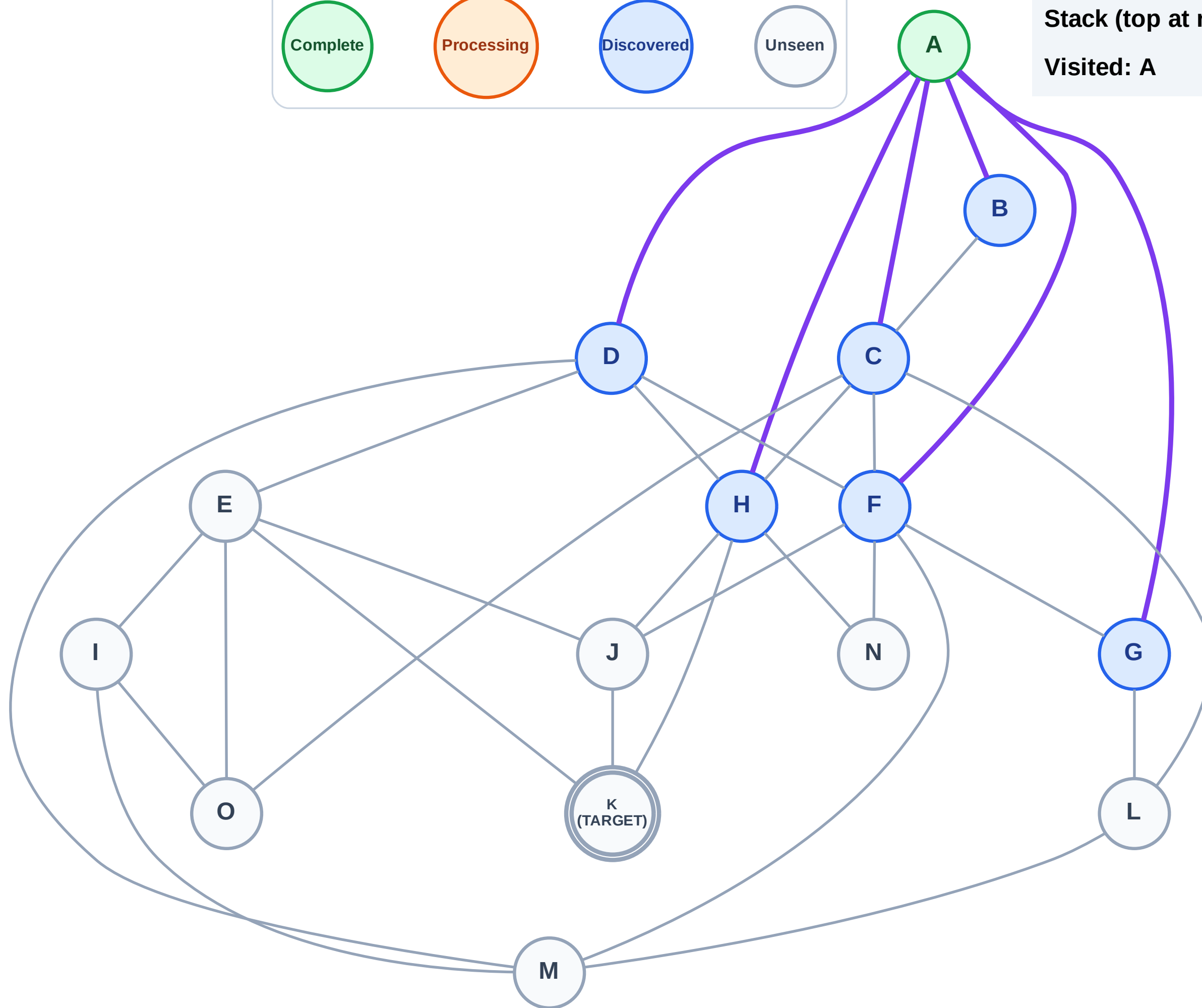
Step 3: Discover H, G, F, D, C, B from A

Legend



Stack (top at right): H | G | F | D | C | B

Visited: A



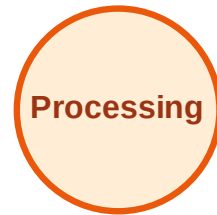
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



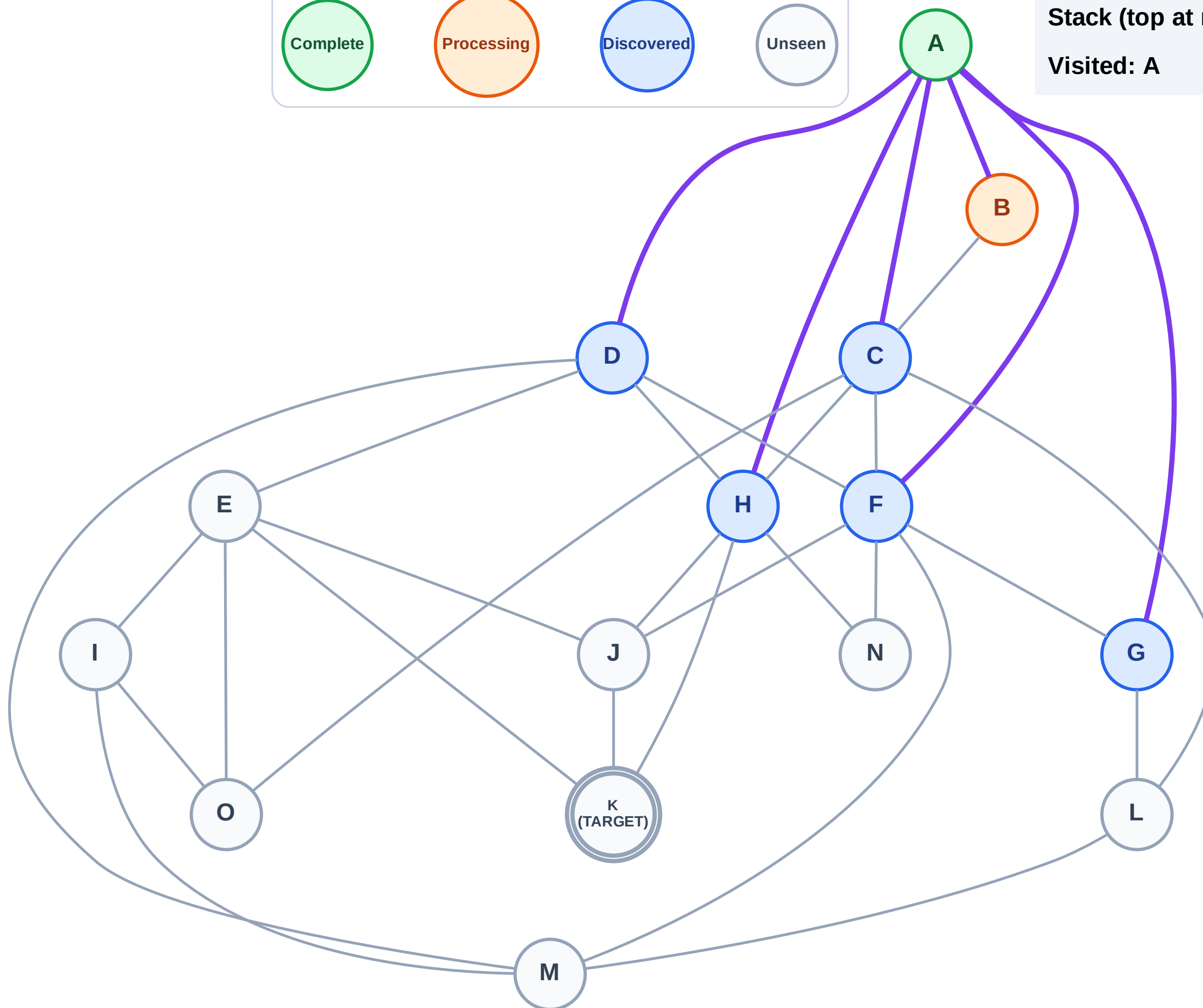
Discovered



Unseen

Stack (top at right): H | G | F | D | C

Visited: A



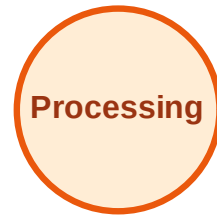
DFS Traversal

Step 5: No new neighbors from B

Legend



Complete



Processing



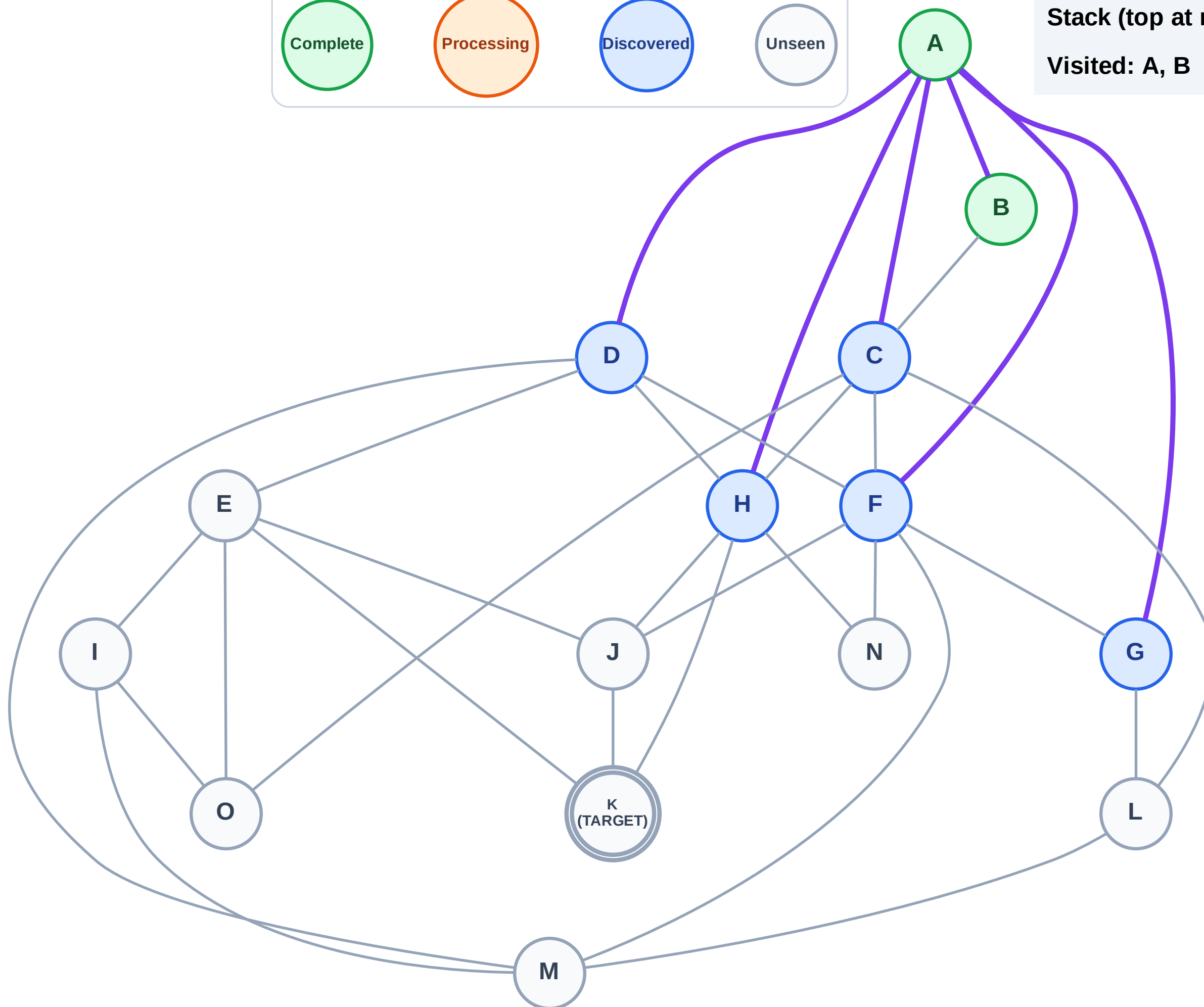
Discovered



Unseen

Stack (top at right): H | G | F | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

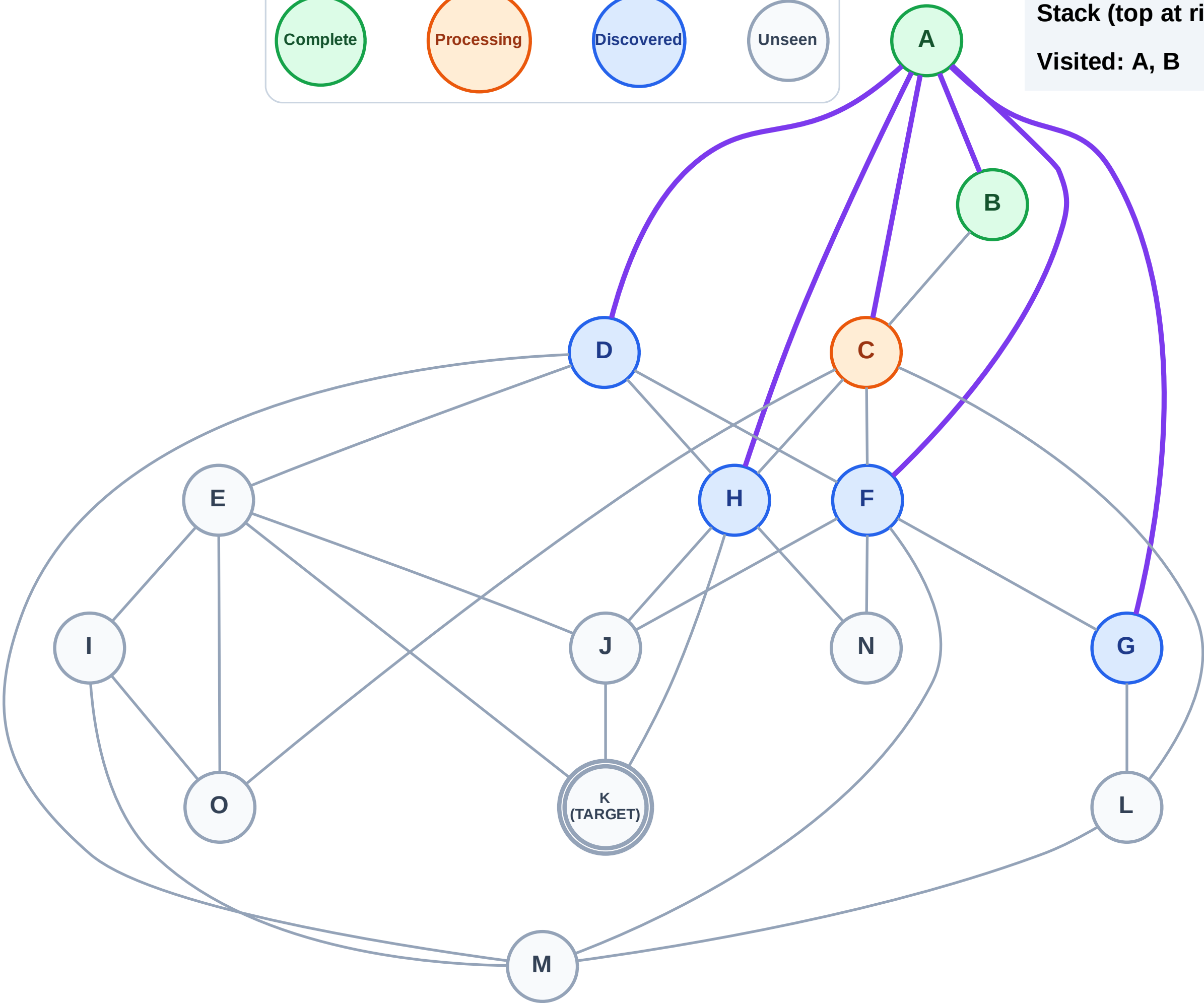
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D

Visited: A, B



DFS Traversal

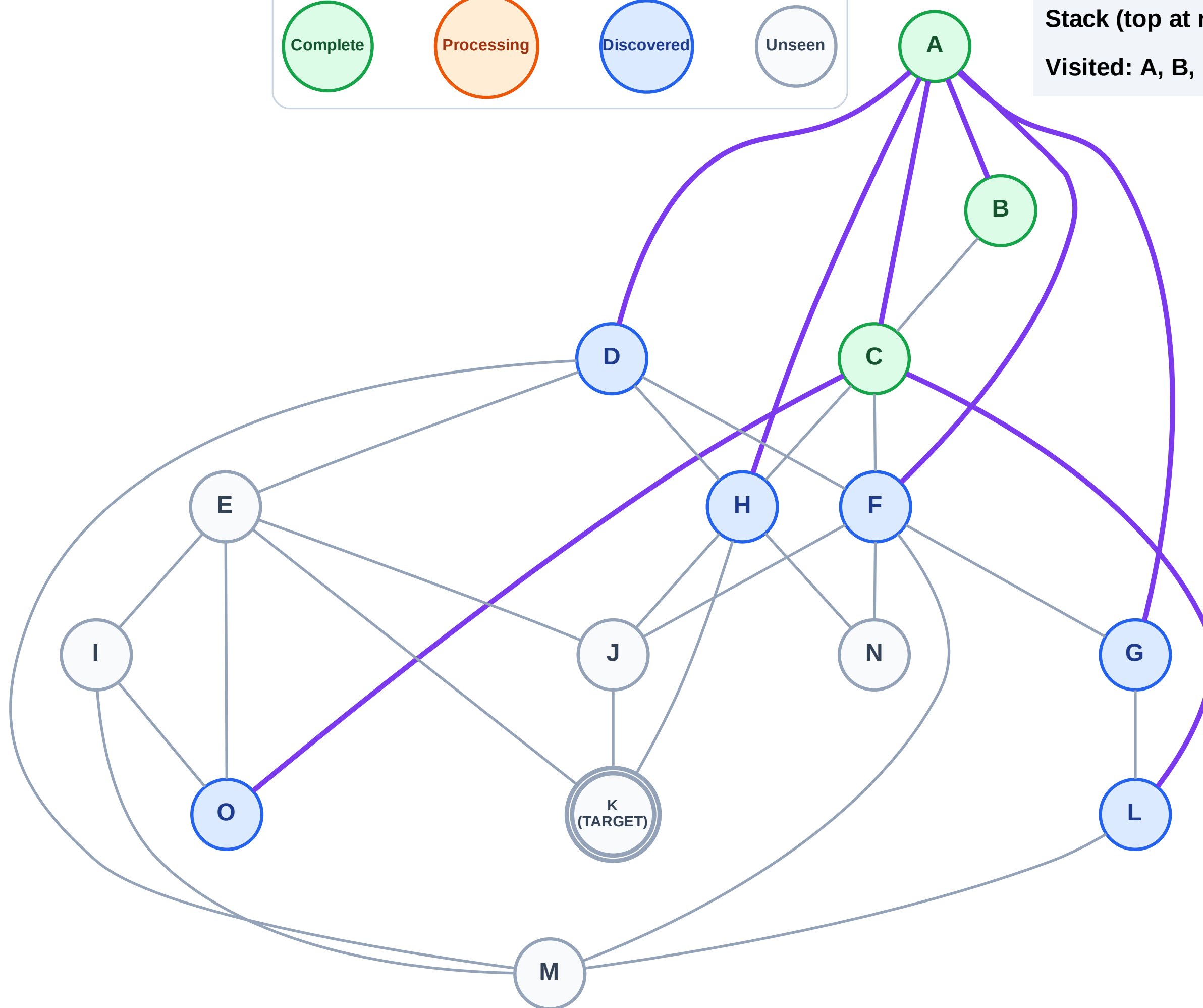
Step 7: Discover O, L from C

Legend



Stack (top at right): H | G | F | D | O | L

Visited: A, B, C



DFS Traversal

Step 8: Process node L

Legend

Complete

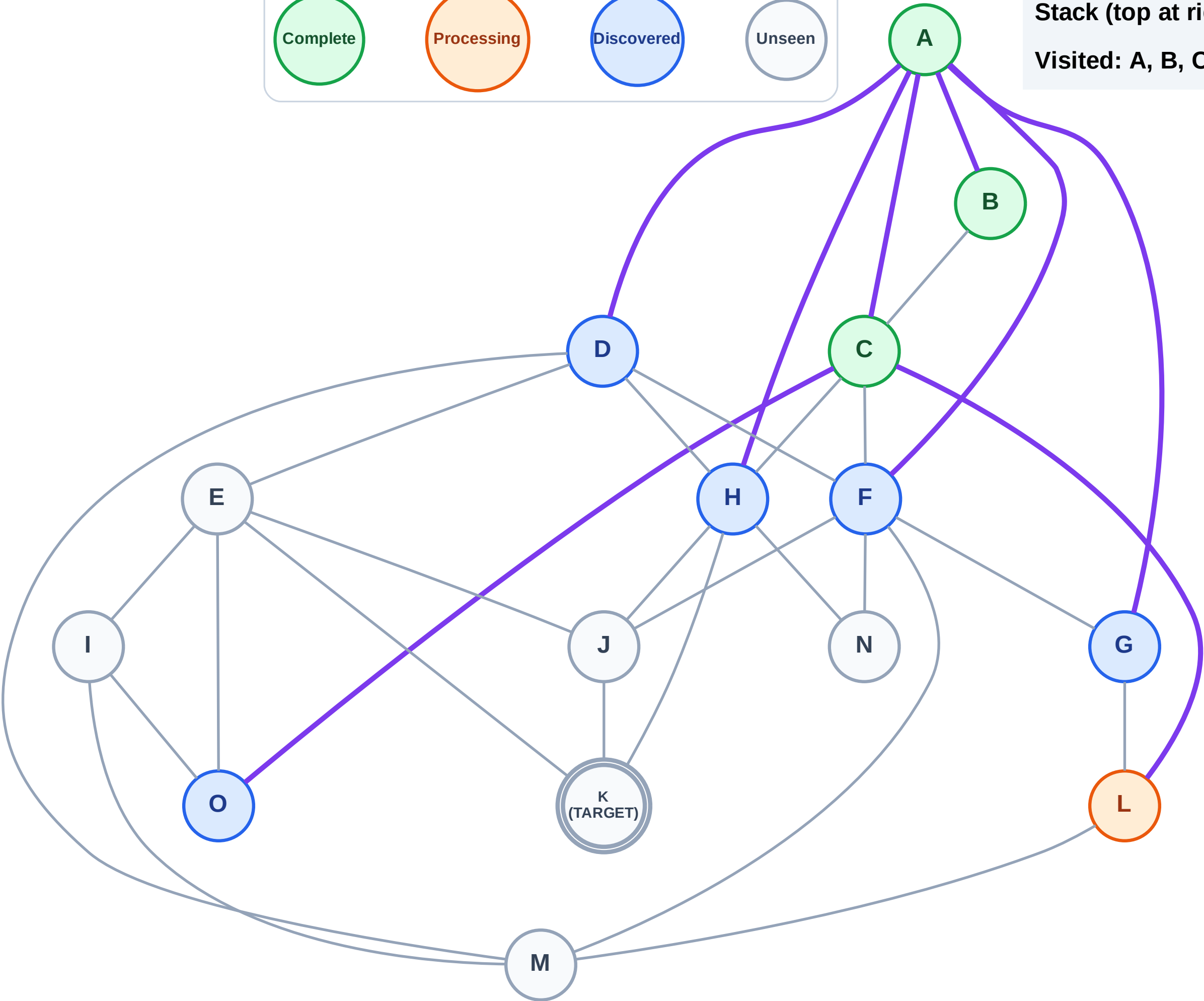
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | O

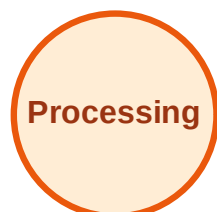
Visited: A, B, C



DFS Traversal

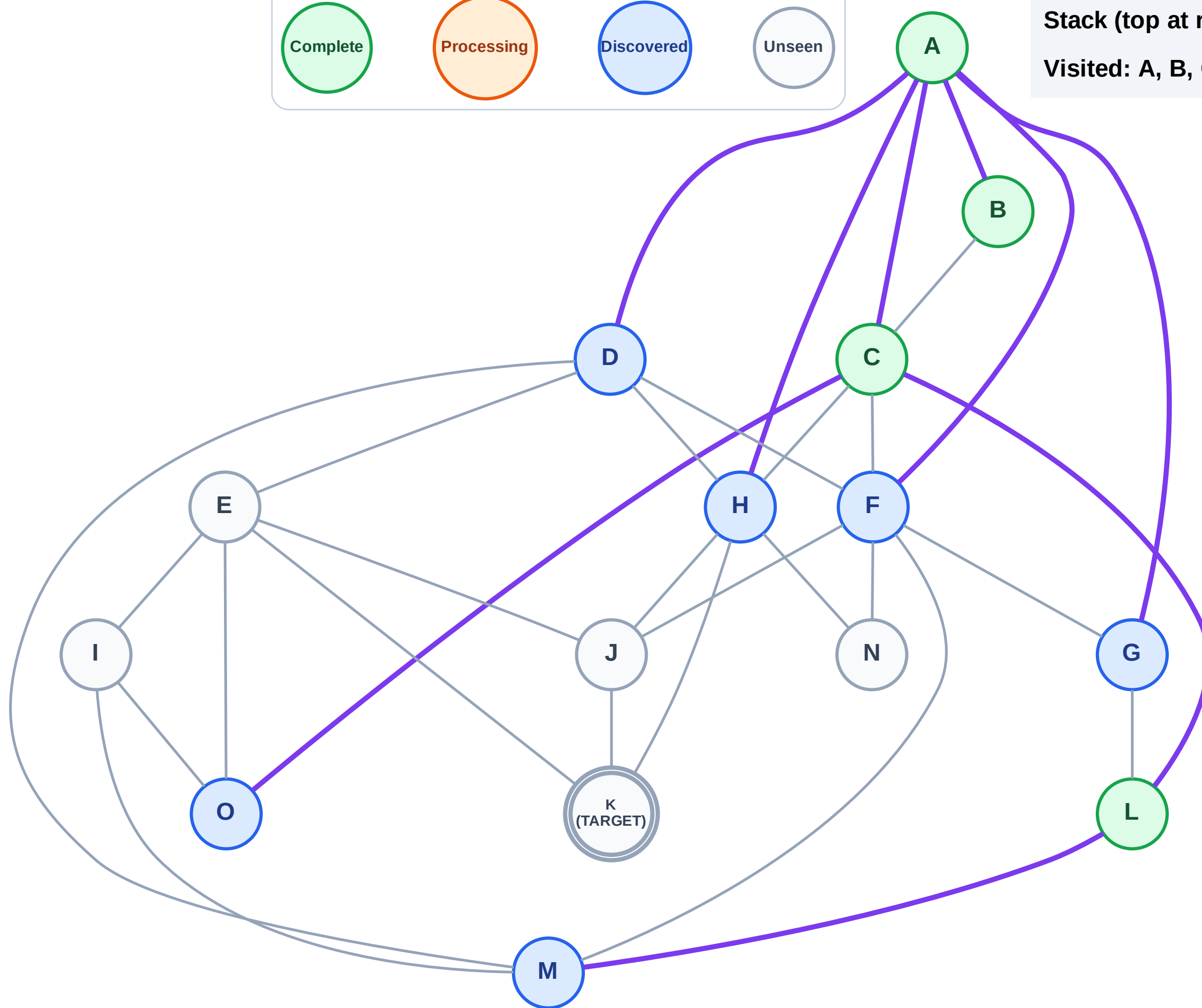
Step 9: Discover M from L

Legend



Stack (top at right): H | G | F | D | O | M

Visited: A, B, C, L



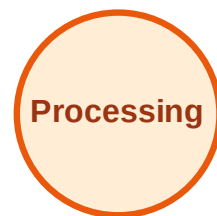
DFS Traversal

Step 10: Process node M

Legend



Complete



Processing



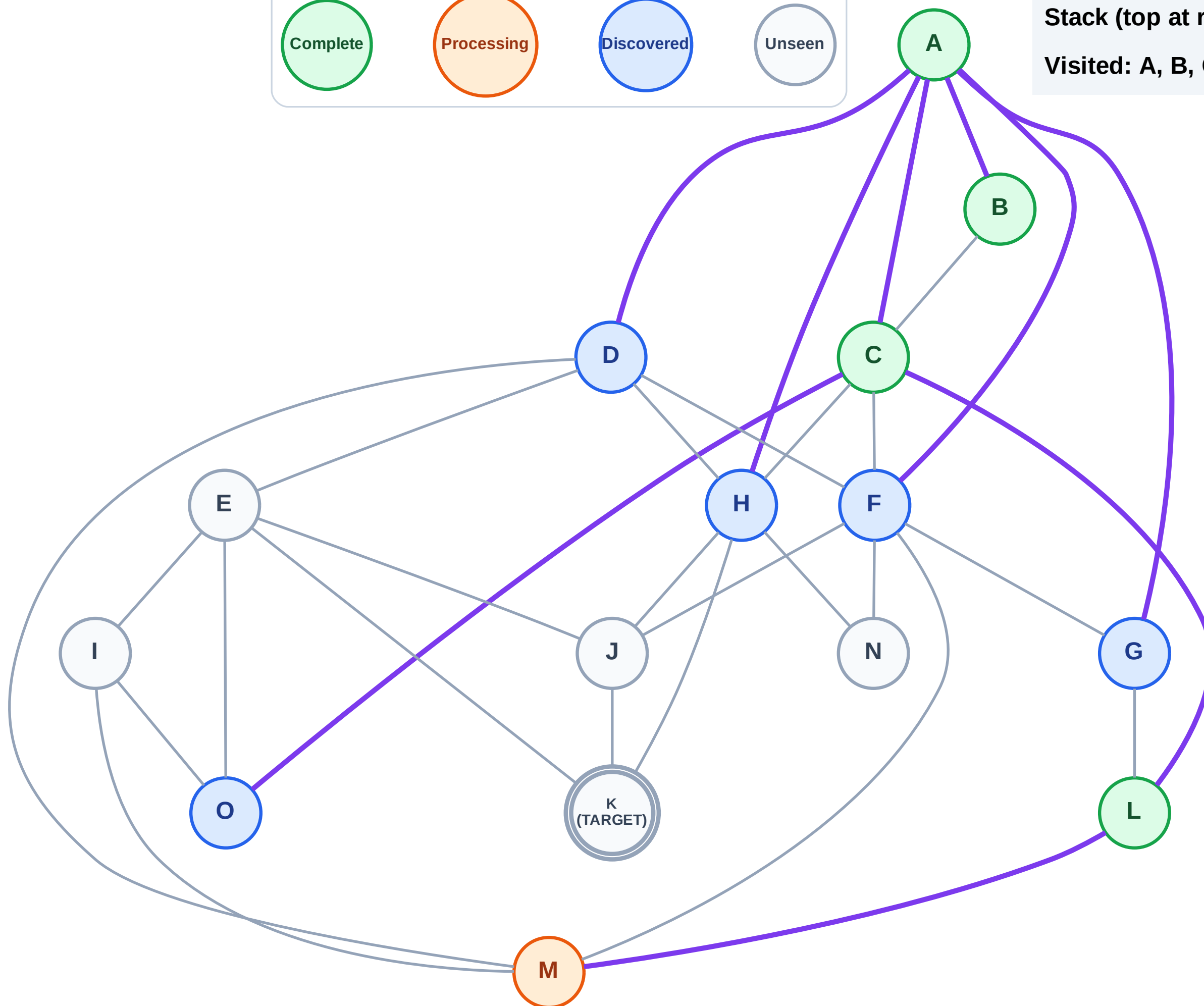
Discovered



Unseen

Stack (top at right): H | G | F | D | O

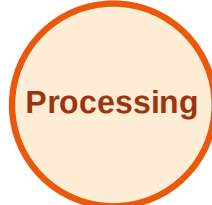
Visited: A, B, C, L



DFS Traversal

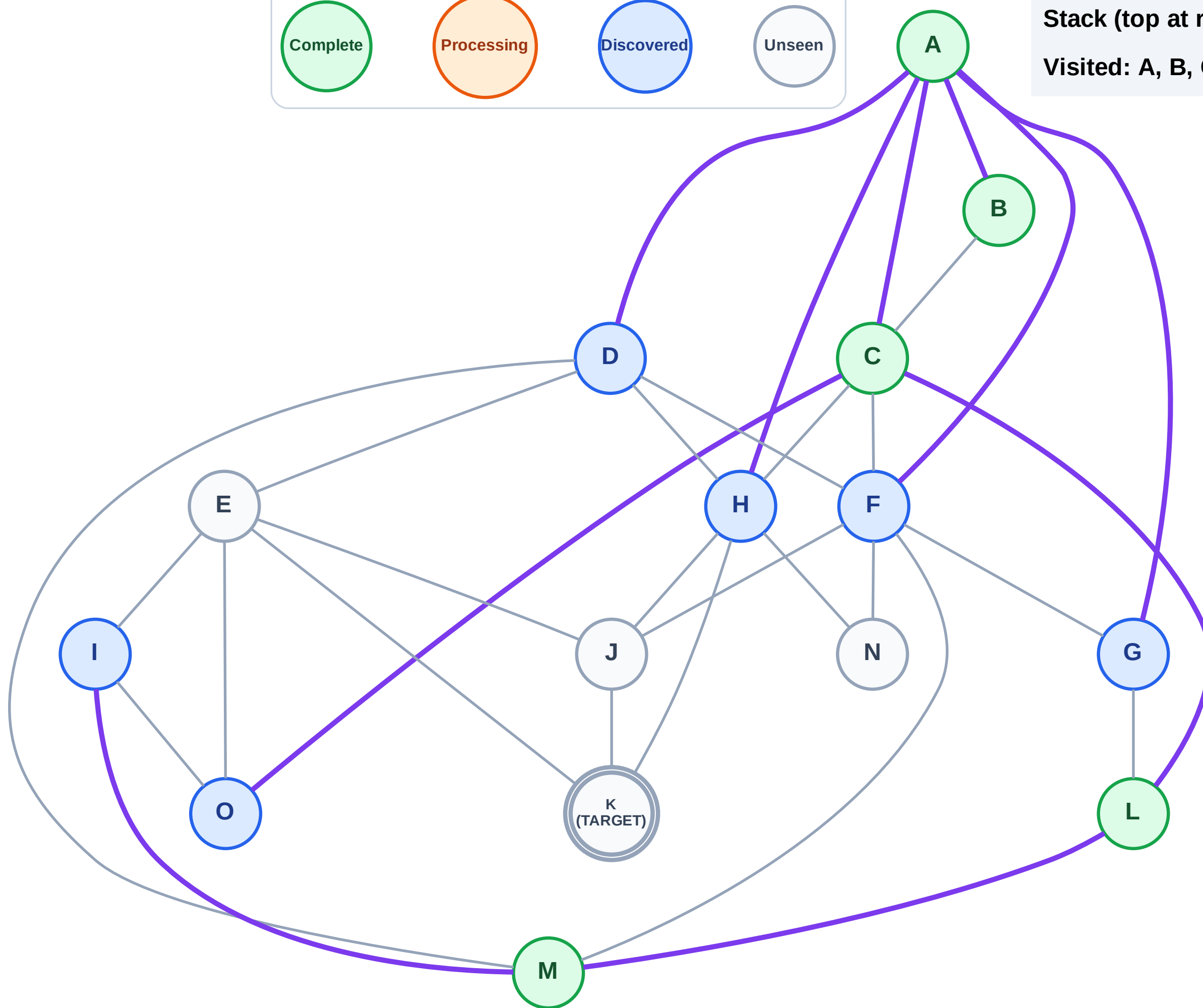
Step 11: Discover I from M

Legend



Stack (top at right): H | G | F | D | O | I

Visited: A, B, C, L, M



DFS Traversal

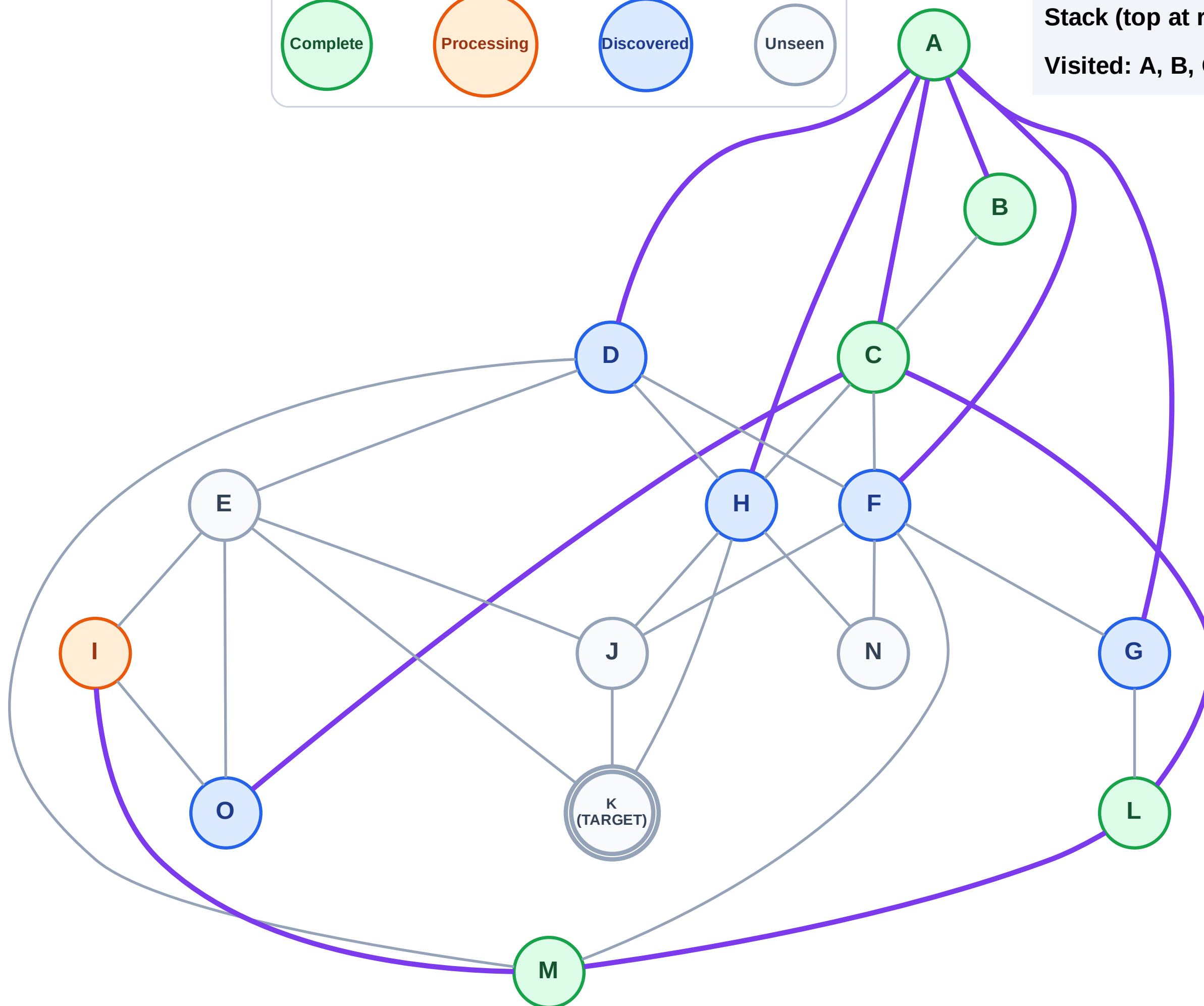
Step 12: Process node I

Legend



Stack (top at right): H | G | F | D | O

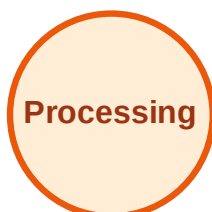
Visited: A, B, C, L, M



DFS Traversal

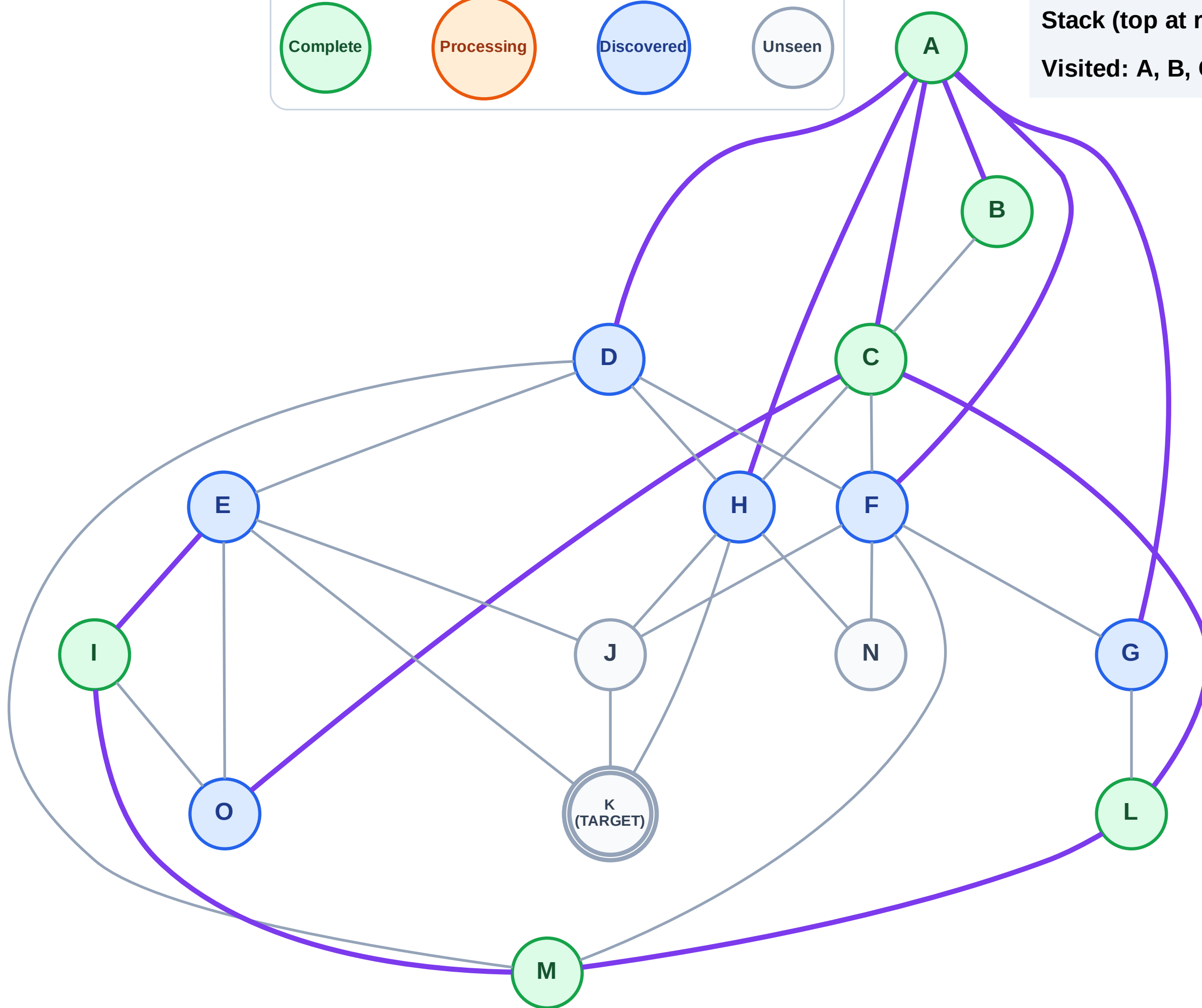
Step 13: Discover E from I

Legend



Stack (top at right): H | G | F | D | O | E

Visited: A, B, C, I, L, M



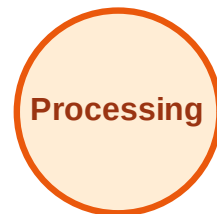
DFS Traversal

Step 14: Process node E

Legend



Complete



Processing



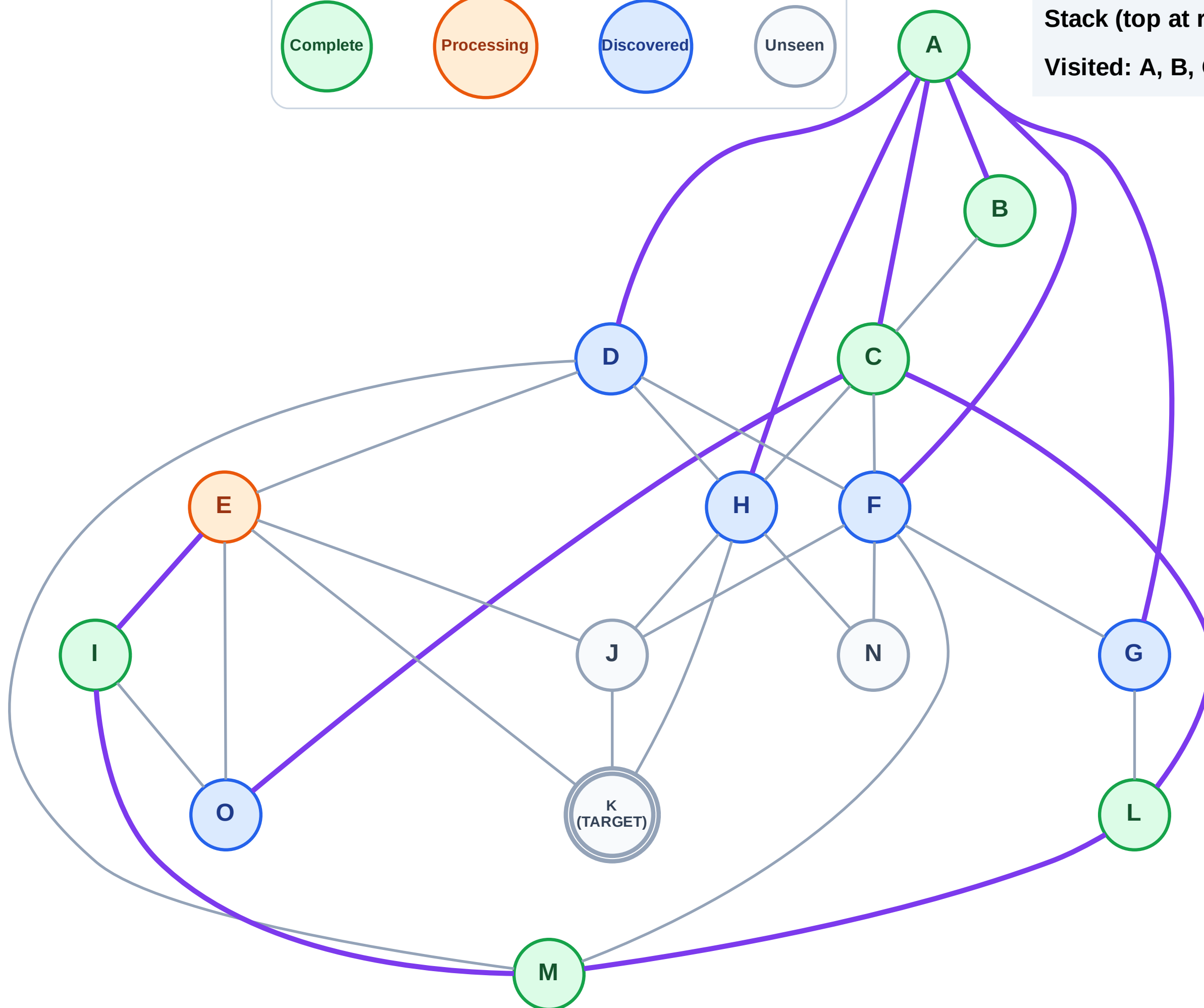
Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, I, L, M



DFS Traversal

Step 15: Discover K, J from E

Legend

Complete

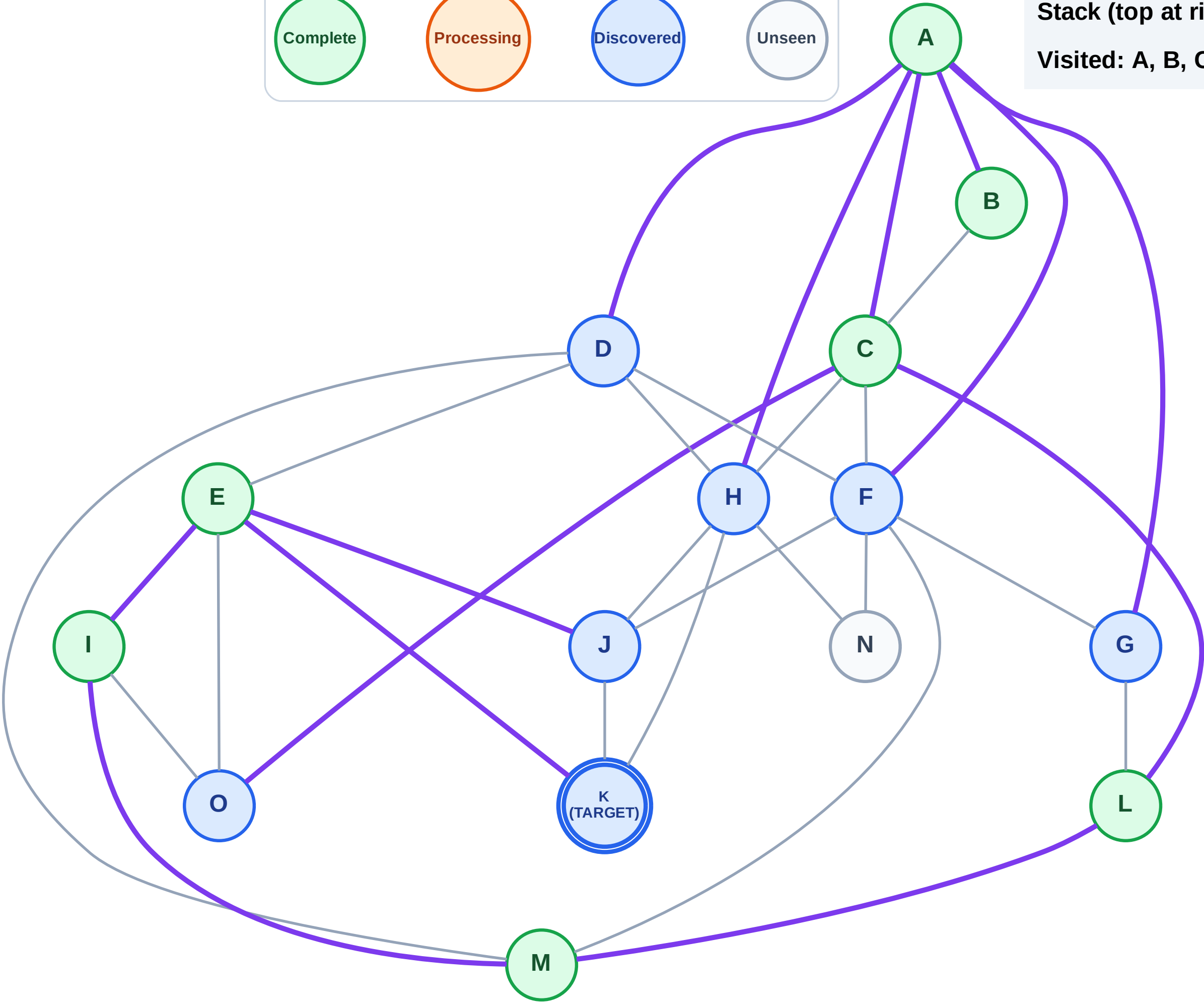
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | O | K | J

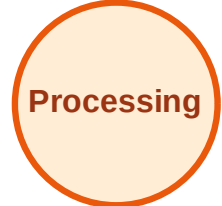
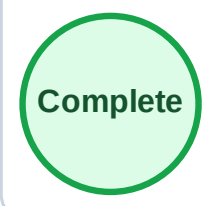
Visited: A, B, C, E, I, L, M



DFS Traversal

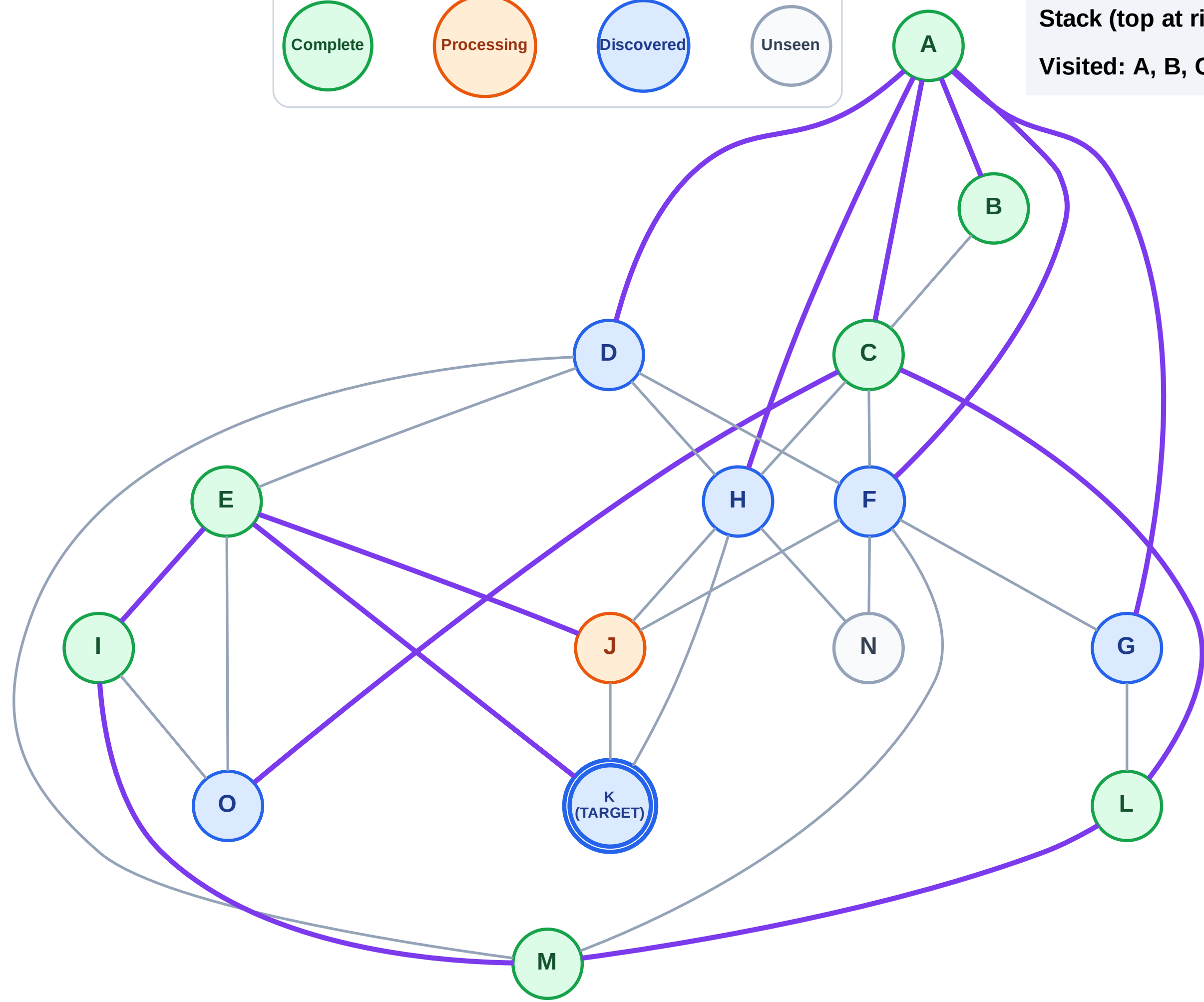
Step 16: Process node J

Legend



Stack (top at right): H | G | F | D | O | K

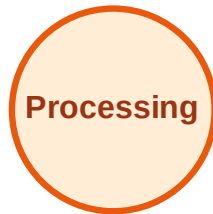
Visited: A, B, C, E, I, L, M



DFS Traversal

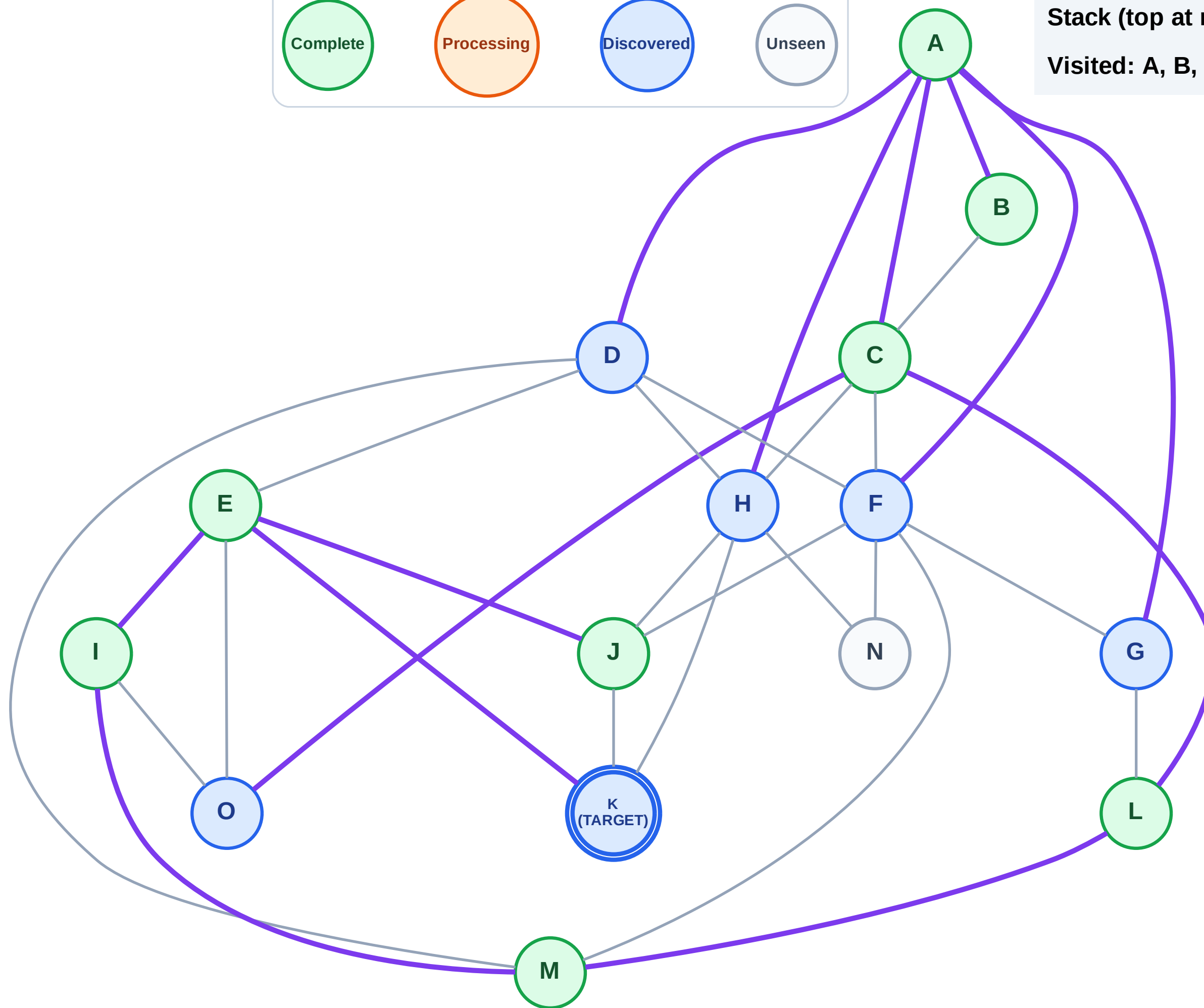
Step 17: No new neighbors from J

Legend



Stack (top at right): H | G | F | D | O | K

Visited: A, B, C, E, I, J, L, M



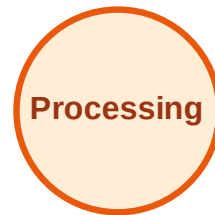
DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



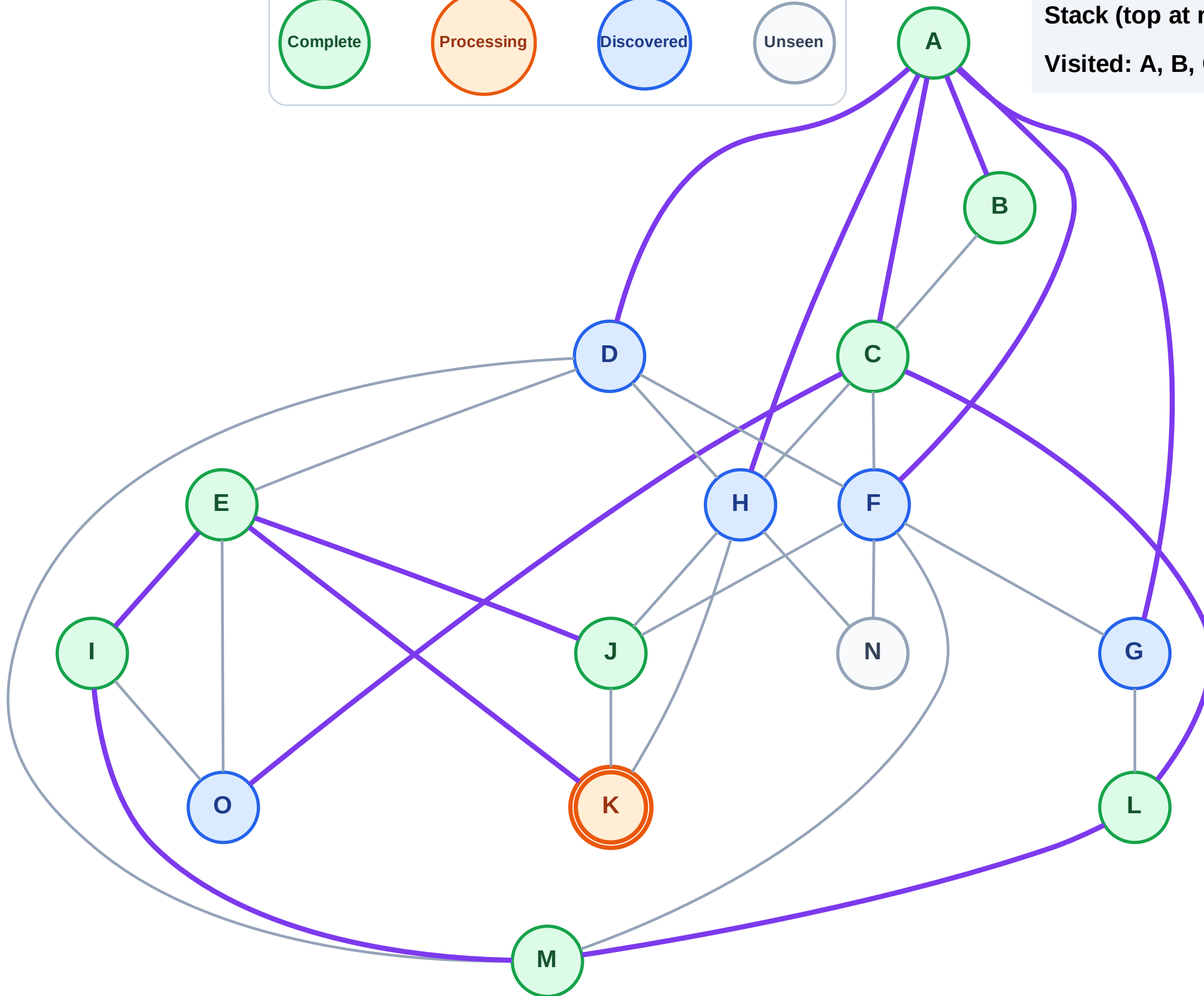
Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, E, I, J, L, M



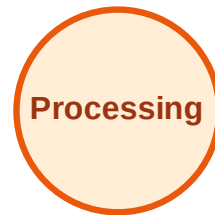
DFS Traversal

Step 19: TARGET NODE K REACHED! Path (6 hops): A → C → L → M → I → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, E, I, J, K, L, M

